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- average output voltage

$$V_o = \frac{1}{T}\int_0^T v(t)\,dt$$

- rms output voltage.

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$$

- ripple factor.

$$RF = \sqrt{(F.F)^2 - 1}$$

- efficiency.

$$\eta = \frac{P_{out}}{P_{in}}$$

- form factor.

$$F.F = \frac{V_{rms}}{V_{dc}}$$

- power factor.

$$PF = \frac{P}{S}$$

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2283

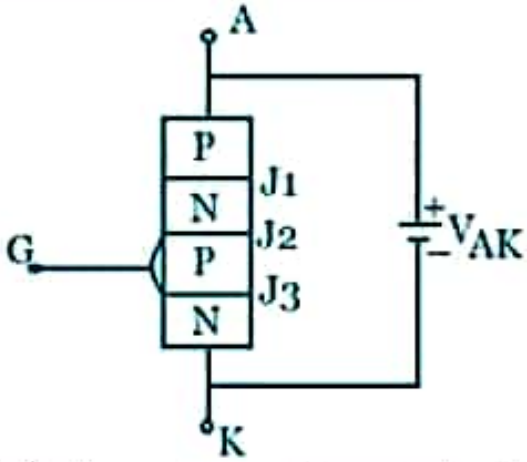
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Student Number:

Question3: ***(10marks)***

3-1 Explain briefly the basic theory of operation of SCR.

The SCR is a four-layer, three-junction and a three-terminal device. The end P-region is the anode, the end N-region is the cathode and the inner P-region is the gate. The anode to cathode is connected in series with the load circuit. Essentially the device is a switch. Ideally it remains off (voltage blocking state), or appears to have an infinite impedance until both the anode and gate terminals have suitable positive voltages with respect to the cathode terminal. The SCR then switches on and current flows and continues to conduct without further gate signals. Ideally the SCR has zero impedance in conduction state. For switching off or reverting to the blocking state, there must be no gate signal and the anode current must be reduced to zero. Current can flow only in one direction.



3-2 State the main features of Firing circuits and write the relationships of firing signals for the full wave controlled rectifier.

Features of firing circuit:

- Firing circuits should produce trigger pulses for the thyristor at appropriate instants.
- There needs to be an electrical isolation between firing circuits and the thyristor. It is achieved using a pulse amplifier or an opto-isolator.

3-3 Explain in detail the turn-off mechanism of an SCR.

To turn OFF the conducting SCR the below conditions must be satisfi

- The anode or forward current of SCR must be reduced to zer below the level of holding current and then.

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- A sufficient reverse voltage must be applied across the SCR to regain its forward blocking state.
- When the anode current decreases below the devices holding current. (I_H) and below this value it automatically turns-"OFF".

3-4 Find the summation of 2 rms values.

$$I_{rms} = \sqrt{I_{rms1}^2 + I_{rms2}^2}$$

3-5 Find the summation of 2 avarage dc values.

$$I_o = I_{o1} + I_{o2}$$

3-6 Write the relationships for the followings:
- Fourier series.

$$v_L(t) = V_o + \sum_{n=1}^{\infty}(a_n Cos\omega t + b_n Sinn\omega t)$$

$$v_L(t) = V_o + \sum_{n=0}^{\infty}(a_n \cos n\omega t + b_n \sin n\omega t)$$

$$V_o = V_{dc} = \frac{1}{T}\int_0^T v_L(t)\,d(\omega t)$$

$$a_n = \frac{2}{T}\int_0^T v(t)\cos n\omega t\,d(\omega t)$$

$$b_n = \frac{2}{T}\int_0^T v(t)\sin n\omega t\,d(\omega t)$$

Or

$$v_o(t) = a_o + \sum_{n=1}^{\infty} c_n \sin(n\omega t + \phi)$$

$$c_n = \sqrt{a_n^2 + b_n^2}$$

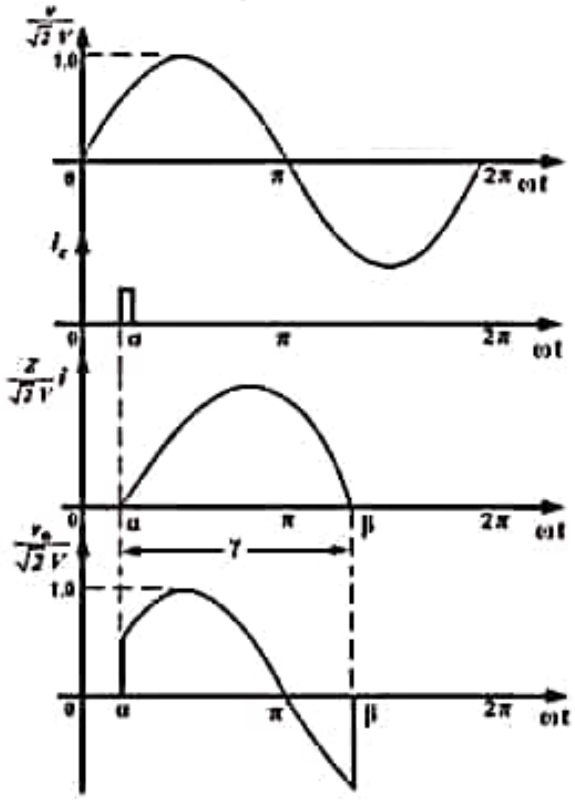
$$\phi_n = \tan^{-1}\frac{a_n}{b_n}$$

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$$v_L + v_R = v_o = v$$

$$L\frac{di}{dt} + Ri = V_m \sin \omega t$$

$$i = \frac{V_m}{Z}\sin(\omega t - \phi) + Ae^{-\frac{Rt}{L}}$$

$$Z = \sqrt{R^2 + (\omega L)^2}$$

$$\phi = \tan^{-1}\frac{\omega L}{R}$$

$$i = 0 = \frac{V_m}{Z}\sin(\alpha - \phi) + Ae^{-\frac{R\alpha}{\omega L}} \Rightarrow$$

$$A = -\frac{V_m}{Z}\sin(\alpha - \phi)\, e^{\frac{R\alpha}{\omega L}}$$

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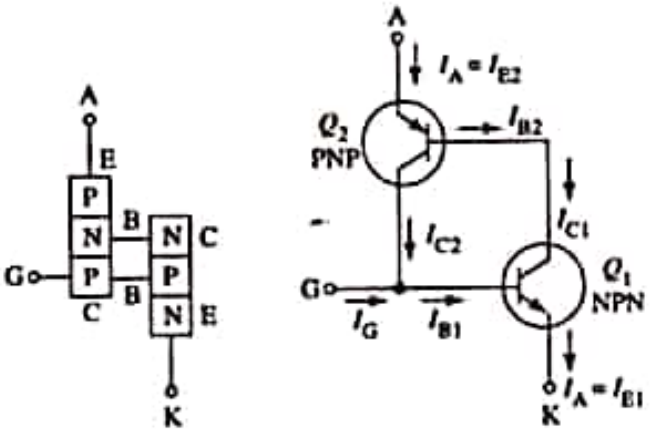
11:00 2283

Albalqa Applied University
Faculty of Engineering Technology

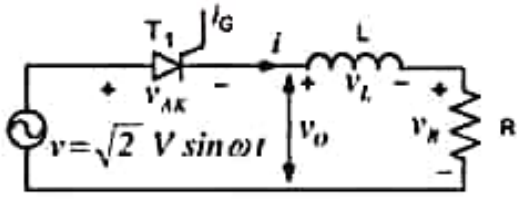
Student Name:
Student Number:

2-3 Explain in detail the Two-Transistor analogy of SCR with proper sketches.

Two Transistor Model Of SCR



2-4 With the help of circuit diagram and necessary waveforms, explain the operation of single phase half -wave controlled rectifier with R-L load and derive the load current for firing angle (α = 30°) and discuss briefly the importance of freewheeling diode in converter circuits.



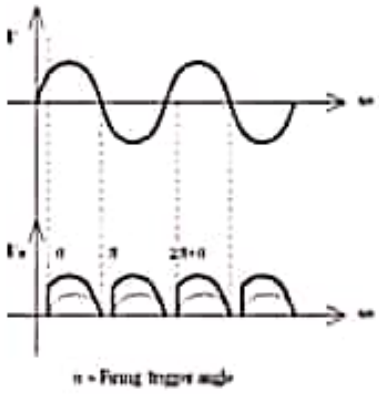
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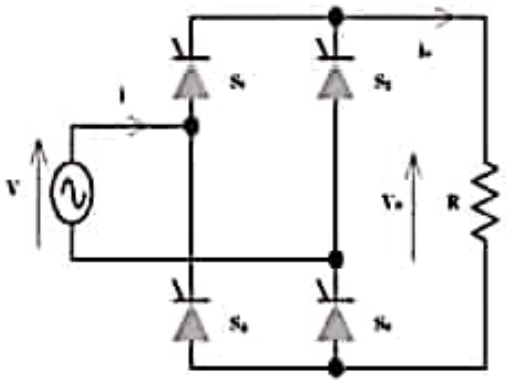
Question2: **(12marks)**

2-1 What is the average value of the dc output voltage of a single-phase half-wave rectified voltage of 311V maximum input voltage value?

$$V_a = \frac{V_m}{\pi} = \frac{311}{\pi} = 99\,V$$

2-2 With the help of circuit diagram and necessary waveforms, explain the operation of single phase full wave controlled rectifier type -bridge feeding R load and derive the expression for load voltage and load current for firing angle (α = 30°).





$$V_a = \frac{2}{2\pi}\int_{\alpha}^{\pi} V_m \, Sin\omega t \, d\omega t$$

$$= \frac{2V_m}{2\pi}\left[-Cos\ \omega t\right]_{\alpha}^{\pi} = \frac{V_m}{\pi}\left[1 + Cos\,\alpha\right]$$

$$I_a = \frac{V_a}{R}$$

$$V_R = \sqrt{\frac{2}{2\pi}\int_{\alpha}^{\pi} V_m^2 \, Sin^2\omega t \, d\omega t} = \frac{V_m}{\sqrt{2}}\sqrt{\frac{1}{\pi}\left[\pi - \alpha + \frac{Sin\,2\alpha}{2}\right]}$$

$$I_R = \frac{V_R}{R}$$

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Question4: (8marks)

Half wave rectifier circuit with switch and RE load circuit, the source voltage v = 311sinωt (v),R= 2Ω and the E= 200(v).If switch is closed during the negative half cycle of the source voltage, calculate:

-The angle at which diode D starts to conduct.

$$\alpha = Sin^{-1}m = Sin^{-1}\frac{V_C}{V_m} = Sin^{-1}\frac{200}{311} = 40^o = 0.697rad$$

-The conduction angle.

$$\gamma = \pi - 2\alpha = \pi - 2\times 0.697 = 1.75\ rad = 100^o$$

-The average value of the load current .

$$I_a = \frac{1}{2\pi}\int_{\alpha}^{\pi-\alpha}\frac{V_m}{R}(Sin\omega t - m)d\omega t$$

$$= \frac{1}{2\pi}\int_{0.697}^{\pi-0.697}\frac{311}{2}\left(\sin\omega t - \frac{200}{311}\right)d(\omega t)$$

- The rms value of the load current.

$$I_R = \sqrt{\frac{1}{2\pi R^2}\int_{\alpha}^{\pi-\alpha}\left[V_m\sin\omega t - V_C\right]^2 d(\omega t)}$$

$$= \sqrt{\frac{1}{2\pi*2^2}\int_{0.697}^{\pi-0.697}(311)^2\left[\sin\omega t - \frac{200}{311}\right]^2 d(\omega t)}$$

-The power delivered by the ac source.

$$P = R\,I_R^2 + V_C \cdot I_a$$

-The power factor at the ac source.

$$Power\ Factor = P.F = \frac{P}{V \cdot I_R}$$

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$$i = \frac{V_m}{Z}\left[Sin(\omega t - \phi) - Sin(\alpha - \phi)\, e^{\frac{R}{L}\left(\frac{\alpha}{\omega} - t\right)} \right]$$

The importance of freewheeling diode in converter circuits:

The SCR will conduct in the negative half cycle for the time of β, therefore the average output voltage decreases due to load inductance.

Avoiding this drawback requires connecting a freewheeling diode **Dm** across the load where the negative portion of the output voltage is eliminated, and the result is keeping the average voltage at the rated value despite of the existing of **L** load.

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Albalqa Applied University
Faculty of Engineering Technology

Student Name:
Student Number:

Mid Exam, Second Semester: 2017/2018
Mechatronics Department

Course Title: Power Electronics
Marks: 40
Lecturers: Prof. Dr.Qazem M. Jaber&Dr.Ghazi M. Qaryouti

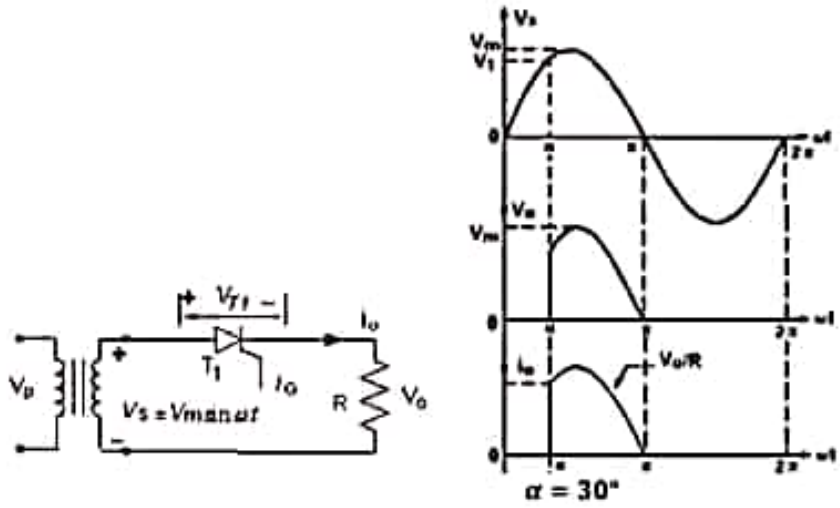
Date: 18/03/2018
Time: 1.25 hour
No. of pages: 4

Question 1: *(10marks)*

1-1 If a 120 V rms voltage is rectified by a single-phase full-wave circuit, calculate the dc output voltage.

$$V_o = V_{dc} = \frac{2}{2\pi}\int_0^{\pi} V_m \, Sin\,\omega t \, d\omega t = \frac{2V_m}{\pi} = \frac{2\sqrt{2}}{\pi} * 120 = 108\,V$$

1-2 A single-phase half-wave controlled rectified with $\alpha = 30°$ with resistive load in what part of the AC cycle does the load current flow?



$$\gamma = \pi - \alpha = 180 - 30 = 150^{o}$$

1-3 Does the SCR conduct if the AK voltage exceeded the forward break over voltage?

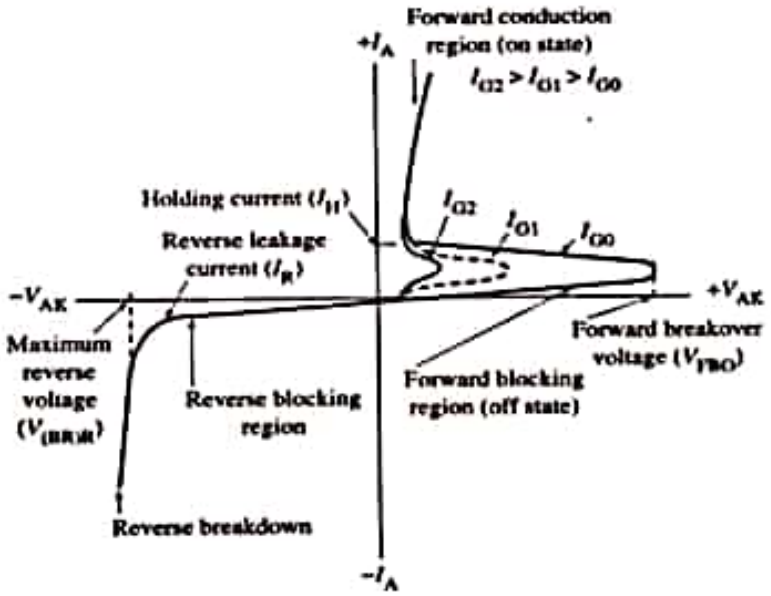
The SCR can be switched on by increasing the forward voltage beyond forward break over voltage. If this voltage exceeded forward voltage the SCR will be damaged.

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Faculty of Engineering Technology

Student Name:
Student Number:



1-4 If an AC Voltage is given by $v(t) = V_{P-P}\,sin\,(\omega t) = 622\,sin\,(314\,t)\ V$.
Calculate the v_{rms} value of this AC voltage.

$$V_{rms} = \frac{622}{2 \cdot \sqrt{2}} = 220\,V$$

1-4 Name the anode minimum input current that can turn on a SCR when the firing signal is removing.

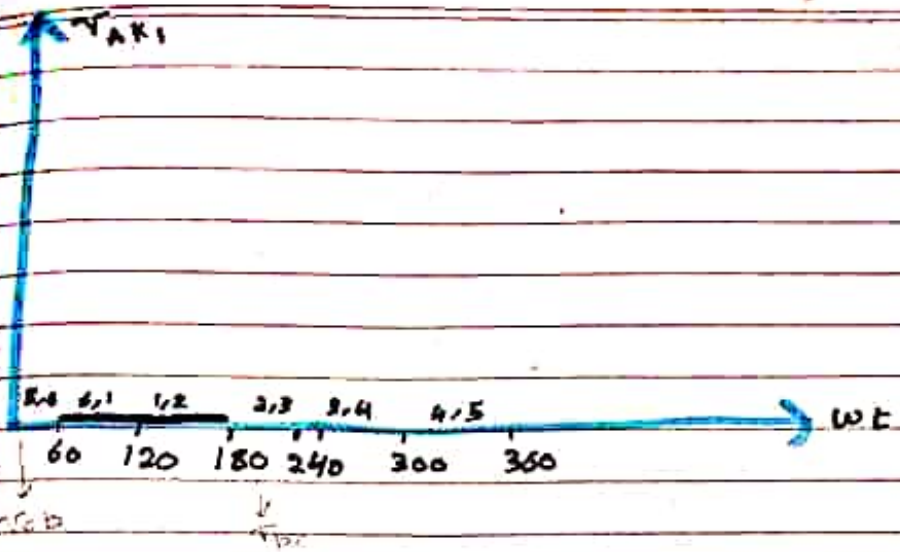
The SCR latching anode current.

1-5 How can you turn off a SCR that is conducting?

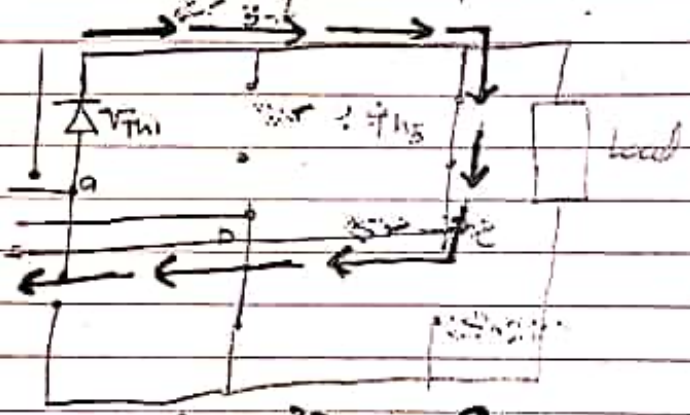
1- التبديل الطبيعي (Natural Commutation).

2- التبديل القسري (Forced Commutation).

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5/6 = Δ loop



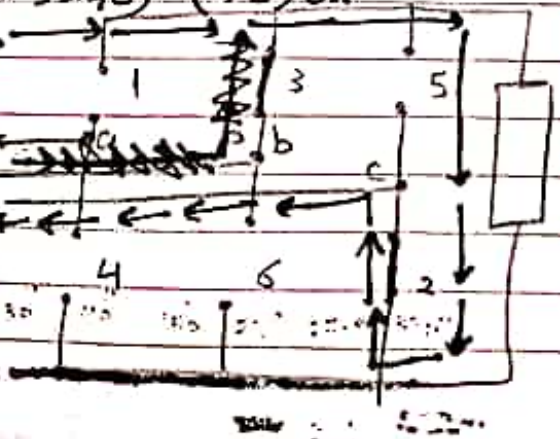
$(0 \rightarrow 60) \xrightarrow{30} (0+30 \rightarrow 60+30) \xrightarrow{90}$

$-V_{an} + V_{TH1} + V_{cn} = 0$

$V_{TH1} = V_{an} - V_{cn}$

$\boxed{V_{TH1} = V_{ac}}$

$(180 \rightarrow 240)$ (2/3) on



$-V_{bn} + V_{TH1}$

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at $\omega t = K \Rightarrow i(t) = 0$

$$i(t) = \frac{V_m \sin(\omega t - \phi)}{R} - \frac{V_m}{R} \sin\phi \, e^{-t\frac{R}{L}}$$

$$0 = \frac{V_m \sin(K - \phi)}{R} - \frac{V_m}{R} \sin\phi \, e^{-\frac{KR}{\omega L}}$$

$\omega t = K$

$$t = \frac{K}{\omega}$$

$i = 0$

$(V_m, \phi, R, \omega, L) \Rightarrow$ قيم معطاة

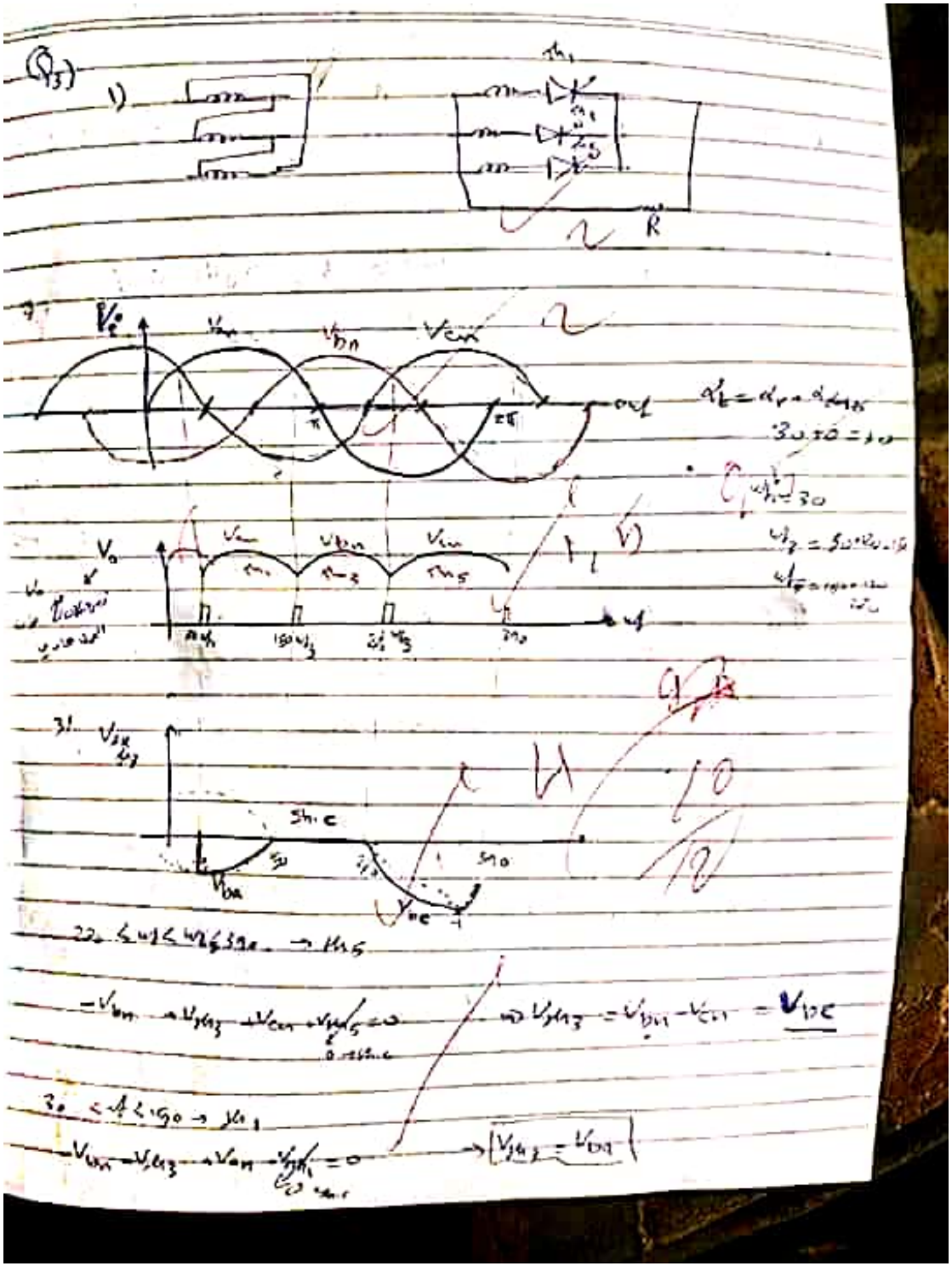
لـ نتج معنا معادلة ϕ بالكلية B,

$$0 = V_m \sin(B - \phi) - \frac{V_m}{R} \sin\phi \, e^{-\frac{BR}{\omega L}}$$

$\omega t = B$

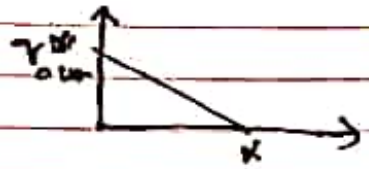
$$t = \frac{B}{\omega}$$

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③ $\alpha = D$ depend on the output voltage:



power factor $\propto V_{out}$

$\propto \frac{1}{\alpha}$

اي انه كلما زادت قيمة α قل ال power factor

$\alpha = 0$	$\alpha = 30^\circ$
power factor is hig higer	power factor is less.

④

three pulse

$I_{rms\ load} = \sqrt{I_{1rms}^2 + I_{3rms}^2 + I_{5rms}^2}$

$= \sqrt{3}\ I_{rms\ th}$

$= \sqrt{3} * 3$

$I_{rms\ load} = 3\sqrt{3}A$

6 pulse

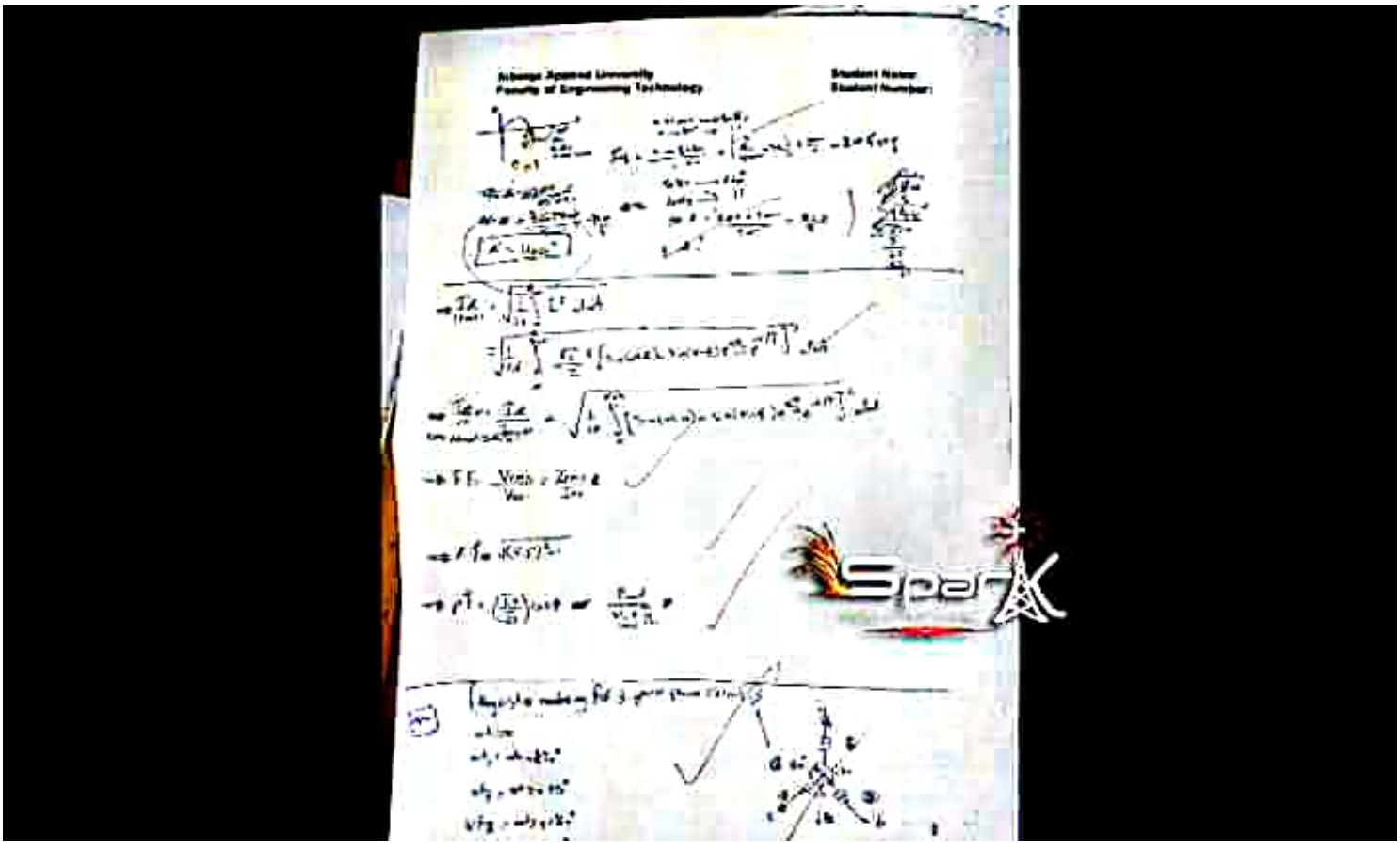
$I_{rms\ load} = \sqrt{I_{1rms}^2 + I_{2rms}^2 + I_{3rms}^2 + I_{4rms}^2 + I_{5rms}^2 + I_{6rms}^2}$.

$= \sqrt{6}\ I_{rms\ th}$

$= \sqrt{6} * 3$

$I_{rms\ load} = 3\sqrt{6}A$

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Question 2

Holding current	Latching current
- associated with turn of process of scR	-associated with turn on process of SCR
-Thyristor can be turned of or returned to Forward bloking only if Farward current falls below a low level current called Holding current	- it is the minimum value of anode current which it must attain during turn on process to maintain conduction when gate pulse is removed.
- Holding current is less than latching Current	-latching current is the 2 to 3 times of holding current.

Half wave	Full wave.
⇒ The tempreture is High because the current is discontinuous and when The transFormer is back to work ⇒ the Inrush current is rise the tempreture of transformer	⇒ The tempreture is less because the current is approximatly continuos and the value of inrush is doe doesn't exsist exsist

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Second Exam... Semester: 2015
Mechatronics Department

Course Title: Power Electronics
Marks:20
Lecturers: Prof. Dr.Qazem M. Jaber

Question 1:

1-Determine the average and rms values in details for the shown in fig.1:

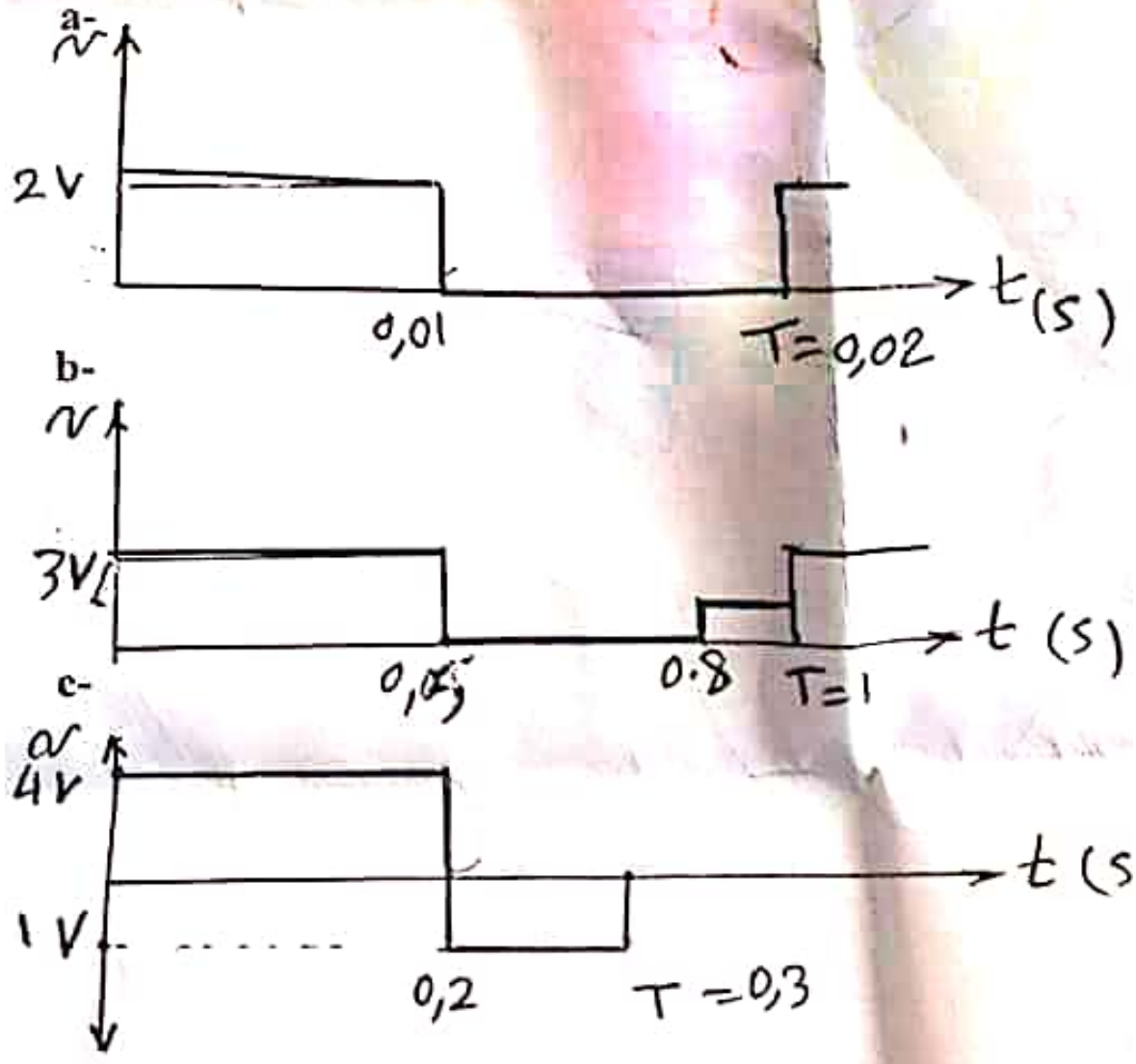
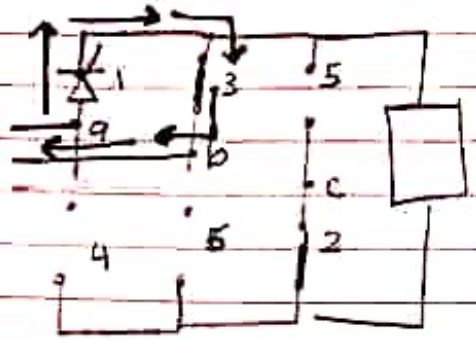


Fig.1

2-A step –down dc chopper has a resistive load of R = 20
400 V. When chopper is ON, its voltage drop is2 V and th
kHz. If the duty cycle is 40%, determine
–Average output voltage
–RMS value of output voltage
–Effective input resistance of chopper
–Chopper efficiency.

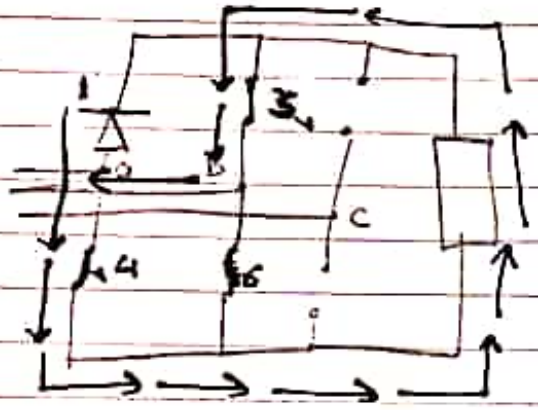
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(180 → 240) ↑ 180+30 ، 240+30
(210 ، 270)



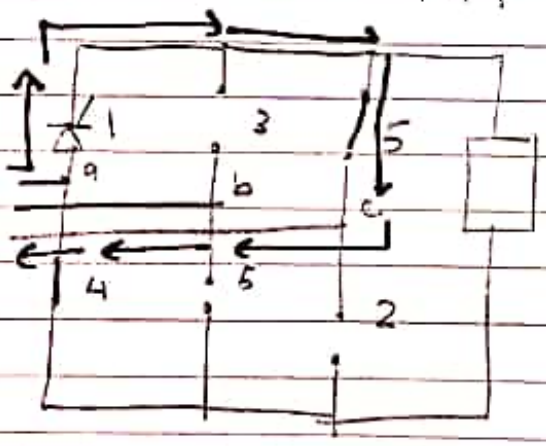
$-V_{an} + V_{Th_1} + V_{bn} = 0$

$V_{Th_1} = V_{ab}$



$-V_{Th_2} + V_{bn} - V_{cn} = 0$

$V_{Th_2} = V_{bc}$

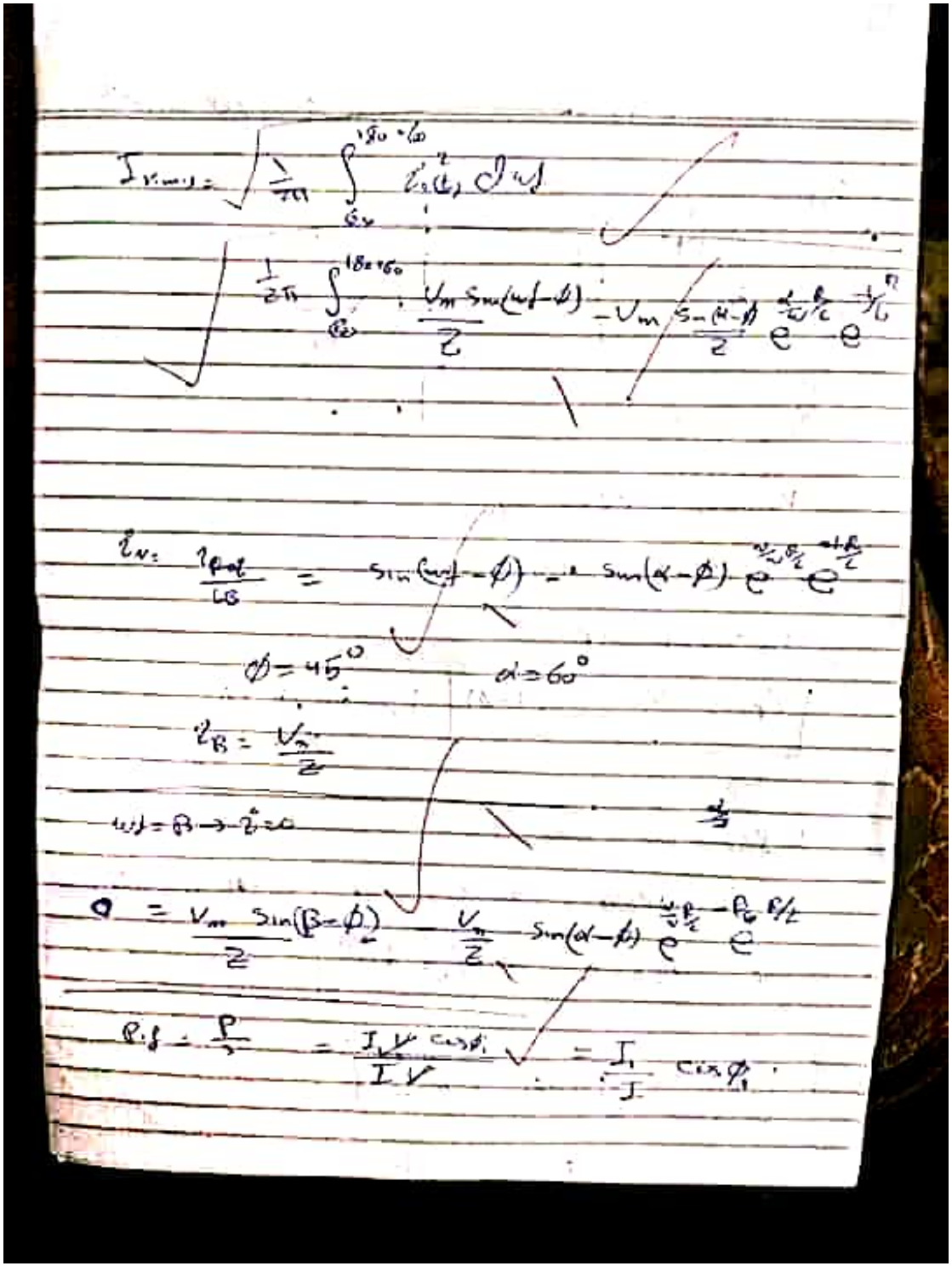


$-V_{Th_3} + V_{cn} - V_{an} = 0$

$V_{Th_3} = V_{ac}$

ونفس الشيء الـ interval اللي بعدها.
باختصار بنوخذ loop بنبلش من الـ ⇐ $(-V_{Th_1})$ وبنكمل
وبنشوف الثايرستادة الشغالة وبنعمل مكانها closed C ⇐
وبنعمل (loop) ⇐

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$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_{\theta_1}^{180+\theta_2} i_2^2(t)\, d\omega t}$$

$$\frac{1}{2\pi}\int_{\theta_2}^{180+\theta_2} \frac{V_m \sin(\omega t-\phi)}{Z} - \frac{V_m \sin(\alpha-\phi)}{Z} e^{\frac{\omega}{2}\frac{R}{L}} e^{-\frac{R}{L}\frac{\pi}{\omega}}$$

$$i_2 = \frac{i_{ph}}{\sqrt{3}} = \sin(\omega t-\phi) - \sin(\alpha-\phi) e^{\frac{\omega}{2}\frac{R}{L}} e^{-\frac{R}{L}}$$

$$\phi = 45^\circ \qquad \alpha = 60^\circ$$

$$i_B = \frac{V_m}{Z}$$

$$\omega t = \beta \rightarrow i_2 = 0$$

$$0 = \frac{V_m}{Z}\sin(\beta-\phi) - \frac{V_m}{Z}\sin(\alpha-\phi) e^{\frac{\omega}{2}\frac{R}{L}} e^{-\beta\frac{R}{L}}$$

$$P.f = \frac{P}{S} = \frac{I_1 V \cos\phi_1}{IV} = \frac{I_1}{I}\cos\phi_1$$

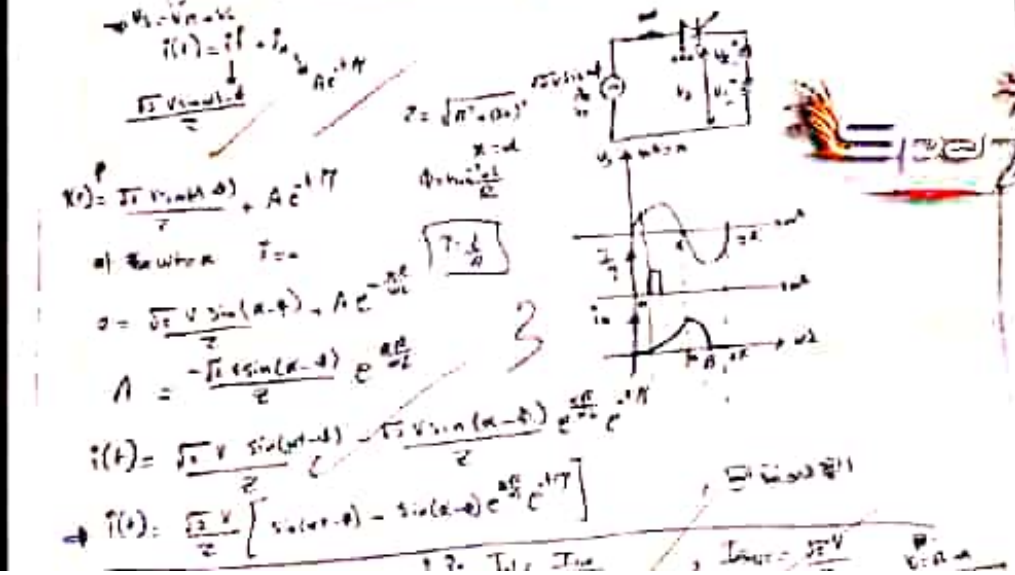
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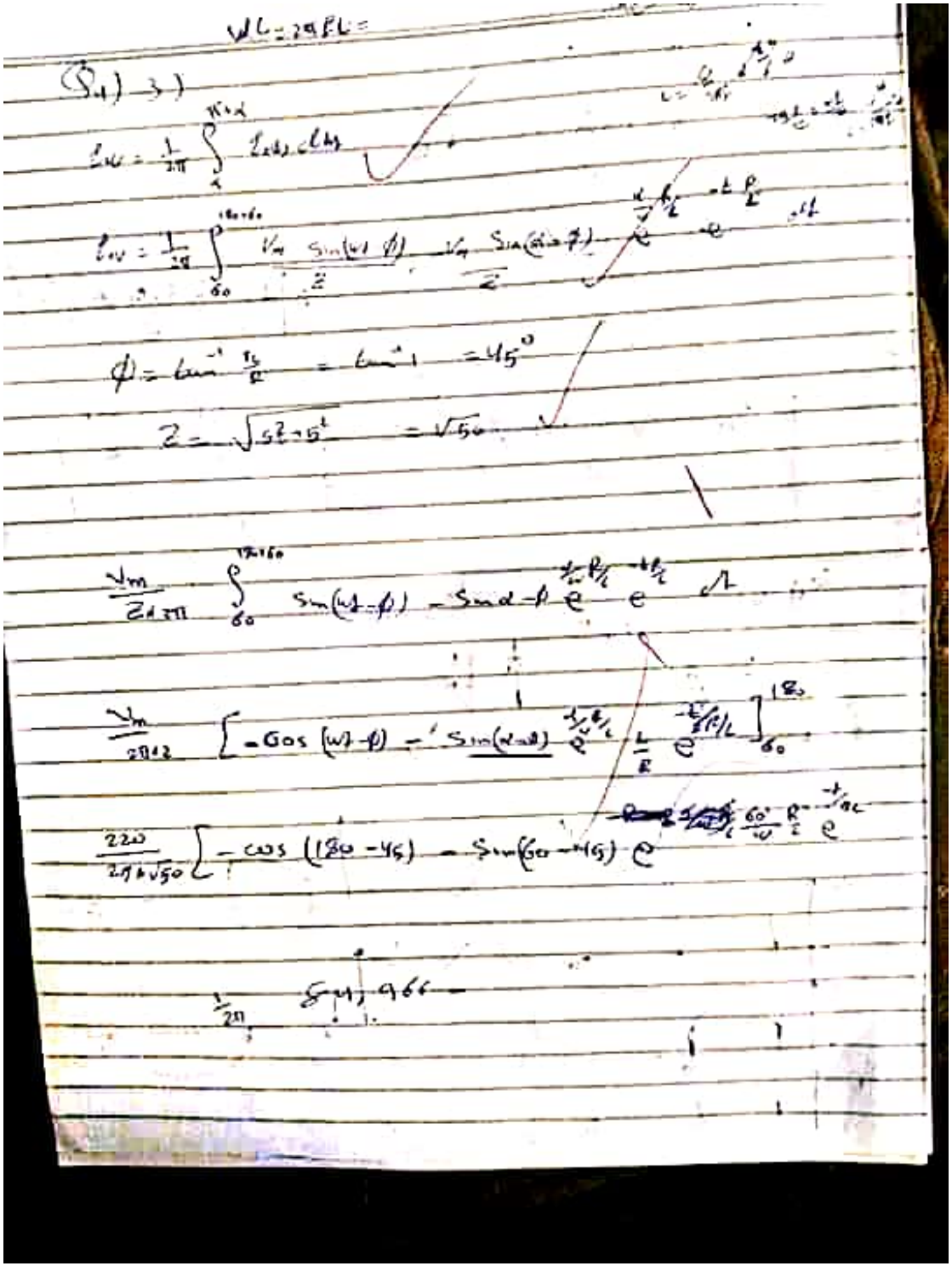
Albalqa Applied University
Faculty of Engineering Technology

Student Name
Student Number

Question2: (3+2+3+1marks)

1-Derive the instantaneous value of the output current for circuit contains AC source ,thyristor and RL load circuit after the the thyristor is triggered at ωt = α .
2-Draw all curves versus time that are used to explaine the quality work of circuit.
3-Write the relation ships of normalized values of average , rms, FF, riple factor and power factor.
4-State the thyristors numbering for three phase power circuit.





$$WL = 2\pi fL =$$

$$(5+j3)$$

$$I_{av} = \frac{1}{2\pi} \int_{\alpha}^{\pi+\alpha} I_m \sin(\omega t - \phi)\, d(\omega t)$$

$$I_{av} = \frac{1}{2\pi} \int_{60}^{180+60} \frac{V_m}{Z} \sin(\omega t - \phi) - \frac{V_m}{Z} \sin(60 - \phi)$$

$$\phi = \tan^{-1}\frac{X_L}{R} = \tan^{-1} 1 = 45^\circ$$

$$Z = \sqrt{5^2 + 5^2} = \sqrt{50}$$

$$\frac{V_m}{2\pi Z} \int_{60}^{180+60} \sin(\omega t - \phi) - \sin\alpha - \phi$$

$$\frac{V_m}{2\pi Z} \left[-\cos(\omega t - \phi) - \sin(\alpha - \phi) \right]_{60}^{180}$$

$$\frac{220}{2\pi\sqrt{50}} \left[-\cos(180 - 45) - \sin(60 - 45) \right]$$

$$\frac{1}{2\pi} \; 8.44 \; 966$$

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Student Name:
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(5marks)

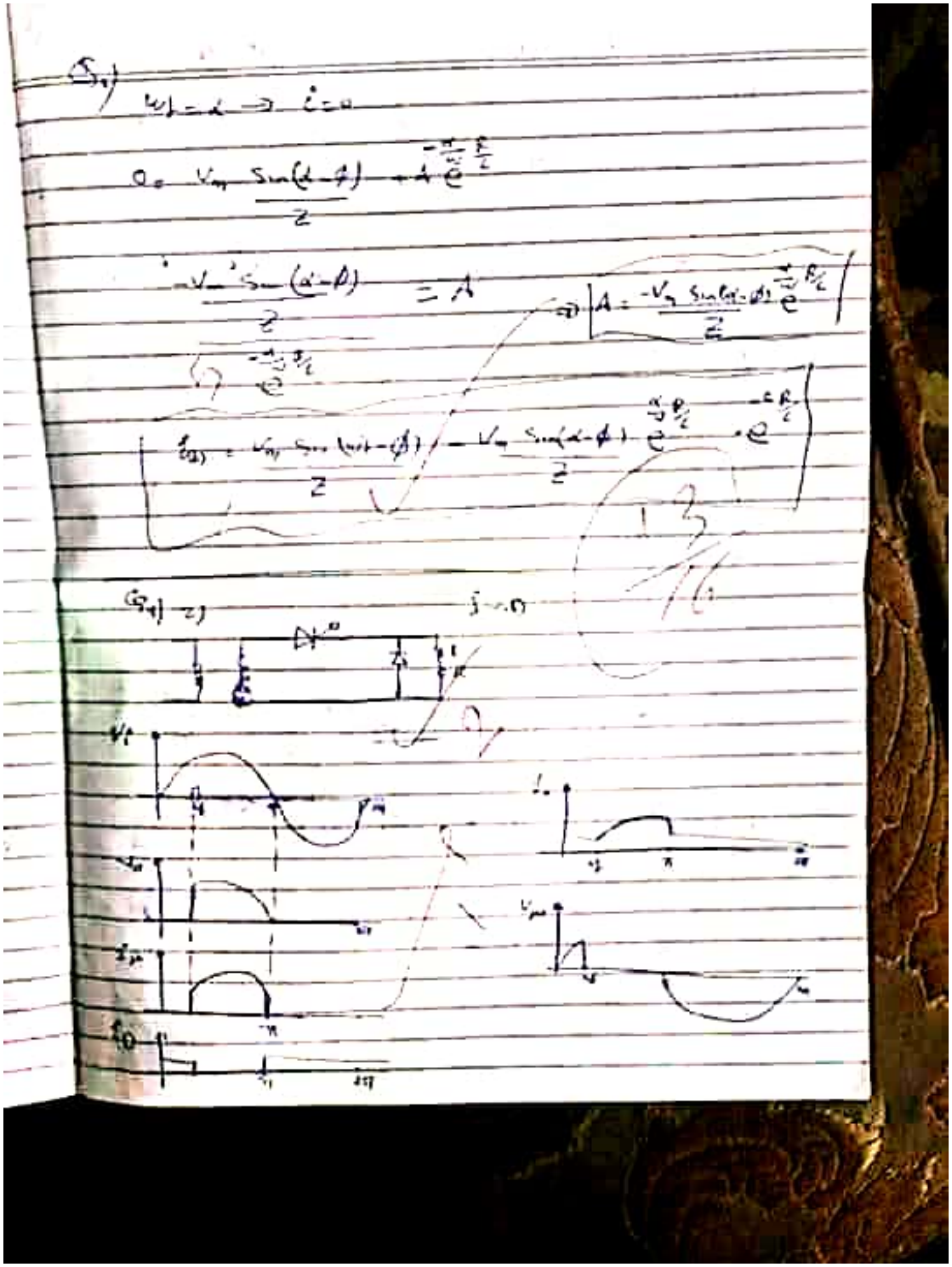
Question4

The singl phase full-wave centre tapped controlled rectifier has the secondary voltage of 110V RMS at 50Hz and a 10 Ω load resistor. The delay angle is $\alpha = 30^{\circ}$ Determine:

1- The average and RMS load current value.
2- The average and RMS thyristor current value.
3- The efficiency
4- The Peak Invers Voltage (PIV) of the thyristor.

Good luck

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Student Name:
Student Number:

Albalqa Applied University
Faculty of Engineering Technology

Med Exam, Second Semester: 2016/2017
Mechatronics Department

Course Title: Power Electronics
Marks: 40
Lecturers: Prof. Dr.Qazem M. Jaber & Dr Ghazi .M. Qaryouti

Date: 21/3/2017
Time: 1:15 hour
No. of pages: 2

Question 1: *(14marks)*

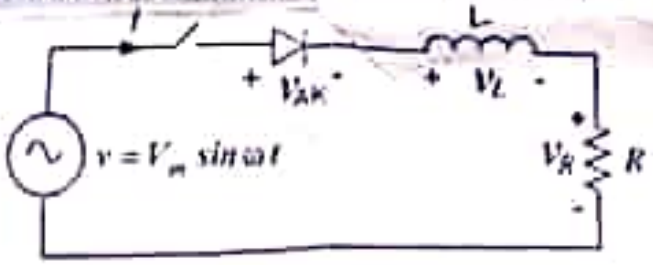
1-Draw the power circuit of three phase controlled rectifier Type Bridge (6 pulses), with ab phase is reference.
2-Draw the line to line waves versus time of power circuit of 6 pulses.
3-Draw the firing signals of thyristores with delay angle (α) = zero.
4-Draw the output instantaneous voltage versus time for circuit of item 1 and item3, with assigning the all necessary number of the thyristor and phase voltage for each interval.
5-Find the anode cathode voltage of thyristot 1 at $\omega t = 30^{\circ}$ and $\omega t = 70^{\circ}$.
6-Scetch the average and the rms voltages the output instantaneous voltage wave versus time.

Question2: *(10marks)*

What is the differences for the following:
1-The lathing and the holding anod currents of thyristor.
2-The temperature of the transformer is fed the half-wave single phase controlled rectifier and the full-wave controlled rectifier with the same resistive load.
3-The power factot of the full -wave single phase controlled rectifier with α = 0°and the full-wave controlled rectifier with α =30° (the load for the two circuits is resistive load).
4-The rms value of the resistive load for the controlird rectifier (3 pulses) and the controlled rectifier (6 pulses)with the rms current of the thyristor is = 3 A.
5-The body of the power thyristor and the the small power thyristor.

Question3: *(8marks)*

Derive the equation of output current that is used to find the value of the extinction angle (β) ,and the average output current (I_o) for the circuit shown in the figure below ,after the switch is closed at t = 0.



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Student Name:
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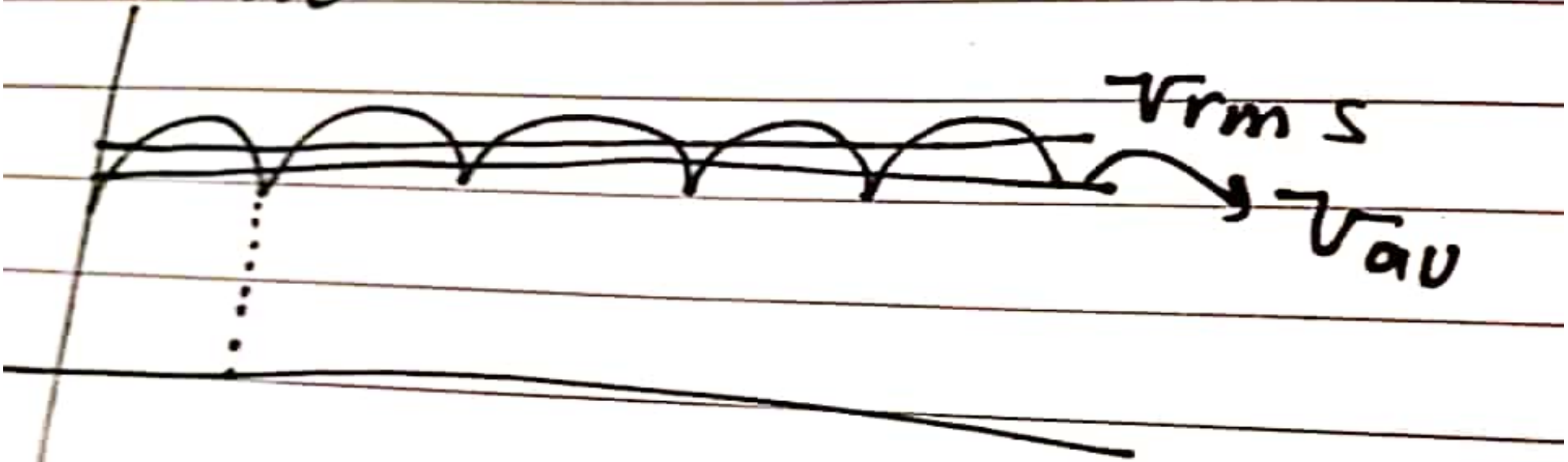
2

Question2: ***(1+1+1+1+3+3marks)***

1-Draw the waves of the source and power circuit for three pulses controlled rectifier with resistive load with delay angle is zero.
2-Draw the firing signals verses time for the circuit in item 2-1 .
3-Draw the out put instantaneous voltage verses time for the circuit in the item 2-1.
4-Draw the third thyristor current verses time for the circuit in the item 2-1.
5-Explain and draw the voltage drop of the third thyristor verses time in the circuit in the item2-1.
6-If the rms and average currents values are 4 amps and 3 amps of the load in the circuit in the item 2-1, select the thyristor.

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Q16



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(Q.1) 2) $i_n \rightarrow$ Source

$0 = V_L + V_R \rightarrow 0 = iR - L\,di/dt$

$\frac{-iR}{L} = \frac{di}{dt} \rightarrow \frac{-i}{L}R\,dt = \frac{di}{i}$

$\frac{-tR}{L} + c = L\dot{i} \rightarrow e^{\frac{-R}{L}t + c} = i_n$

$\frac{c}{A} e^{\frac{R}{L}t} = i_n \rightarrow \boxed{i_n = A e^{\frac{-R}{L}t}}$

$i_f = \frac{V_s}{R}$

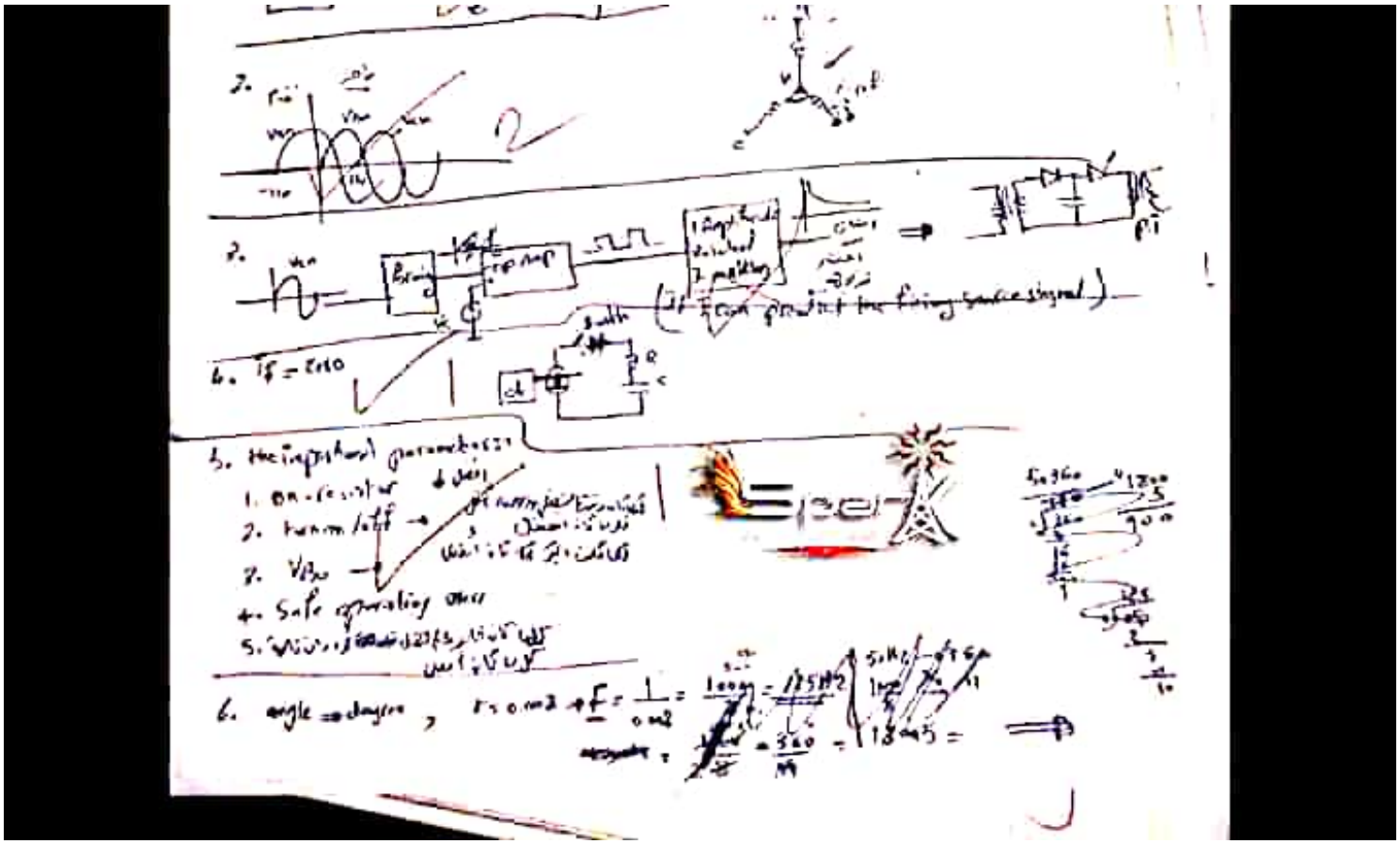
$i_n = \frac{V_s}{R} + A e^{\frac{-R}{L}t}$

$e \to \infty \Rightarrow A \rightarrow$ at $t=0 \rightarrow i=0$

$0 = \frac{V_s}{R} + A \rightarrow \boxed{A = \frac{-V_s}{R}}$

$\boxed{i_f = \frac{V_s}{R} - \frac{V_s}{R} e^{\frac{-R}{L}t}}$

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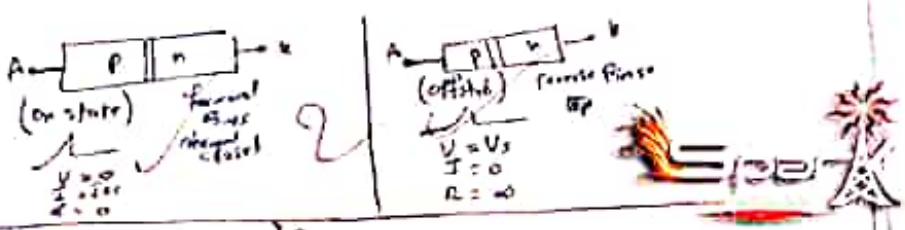
Albalqa Applied University
Faculty of Engineering Technology

Student Name:
Student Number:

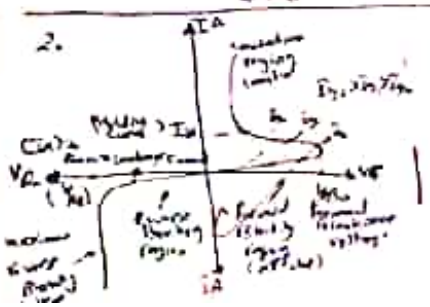
Question3. (Total of 5 marks)

1-Represent electrically the P-N junction on state and the P-N junction off state and find their values.
2- Draw the real volt- amp characteristic of thyristor with important ratings.
3-Define the suitable firing signal and latch current
4-Find the average and the arms values of pulse with duration time is 0.001(s), the period is 0.01(s) and the high of pulse is 2 v.
5-Draw the symbol of thyristor and explain how to determine the terminals of thyristor.

1.



2.



3. Suitable firing signal

Latch Current: the minimum anode current must flow through the SCR in order to stay initially after the gate signal removed

Q4)

$$V_{av} = \frac{1}{2\pi}\int_{30}^{\pi} V_m \sin\omega t \, d\omega t$$

$$= \frac{\sqrt{2}}{2\pi} V_{rms(s)} \cos\omega t \Big]_{30}^{\pi}$$

$$= \frac{-\sqrt{2}}{2\pi} * \frac{220}{110}\left[-1-\frac{\sqrt{3}}{2}\right]$$

$$V_{av} = +\frac{\sqrt{2}}{4\pi}\frac{220}{110}\left[1+\frac{\sqrt{3}}{2}\right]$$

$$I_{av} = \frac{V_{av}}{R}$$

$$I_{av} = \frac{\frac{\sqrt{2}}{2\pi}(110)\left(1+\frac{\sqrt{3}}{2}\right)}{10} \text{ A}$$

$$I_{rms} = \left(\sqrt{\frac{1}{2\pi}\int_{\alpha}^{\pi} V_m^2 \sin^2\omega t \, d\omega t}\right)/R$$

$$= \left(\sqrt{\frac{V_m^2}{2\pi}\int_{\alpha}^{\pi}\frac{1}{2}(1-\cos 2\omega t)\, d\omega t}\right)/R$$

$$= \left(\sqrt{\frac{V_m^2}{4\pi}\left(\omega t - \frac{\sin 2\omega t}{2}\right)_{\alpha}^{\pi}}\right)/R$$

$$= \left(\frac{V_m}{2}\sqrt{\frac{1}{\pi}(\pi-\alpha)-\left(0-\frac{\sin 2\alpha}{2}\right)}\right)/R$$

$$= \frac{V_m}{2}\sqrt{\left(\frac{1-\alpha}{\pi}+\frac{\sin 2\alpha}{2}\right)} \rightarrow \text{rad}$$

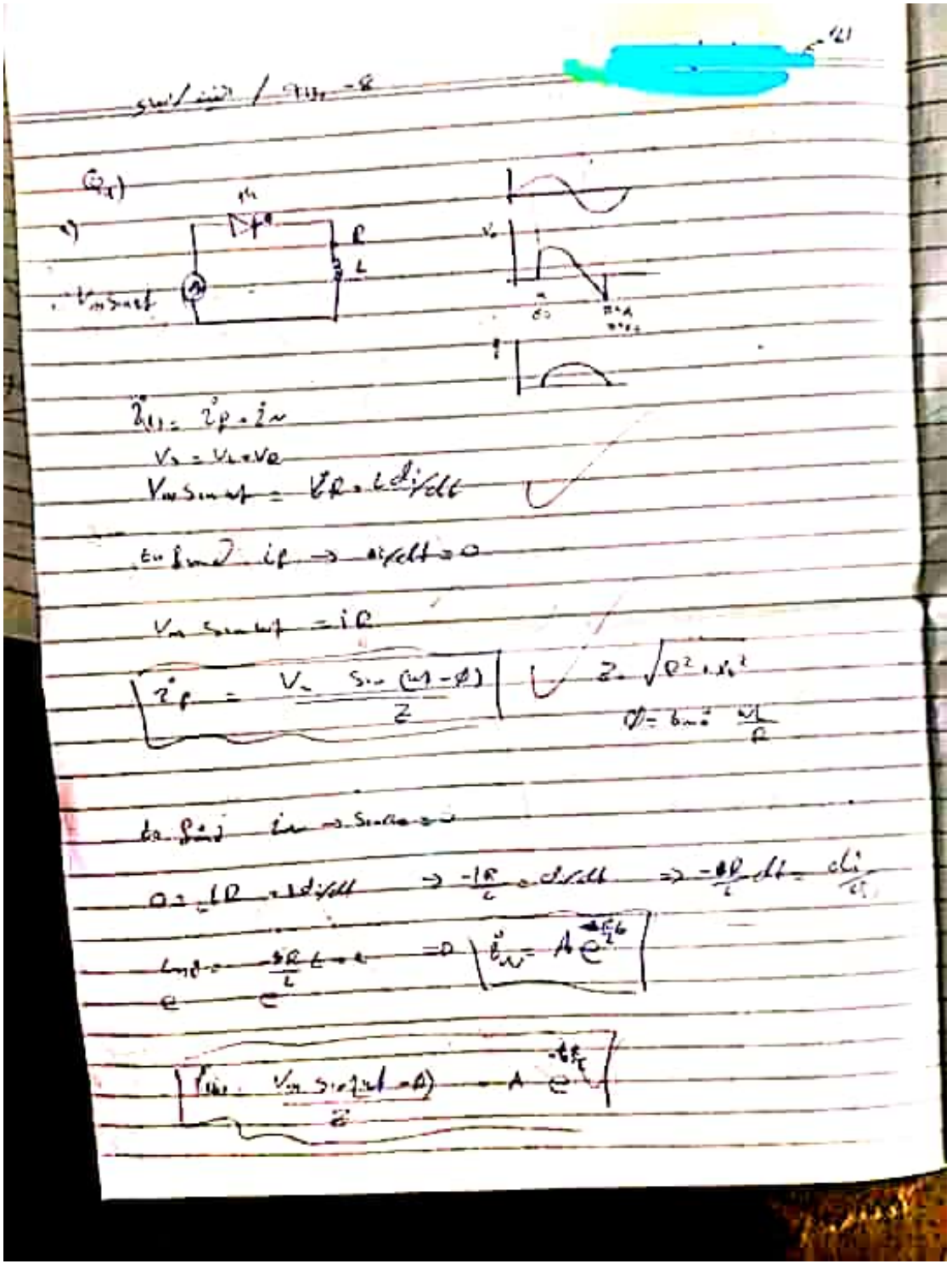
$$= \left(\frac{\sqrt{2}(110)}{2}\sqrt{1-0.16+\frac{\sin 60^\circ}{2}}\right)/10$$

② $I_{av\,Th} = \frac{I_{av\,Load}}{2}$

$I_{rms\,Th} = \frac{I_{rms\,Load}}{\sqrt{2}}$

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Q1)

1)



$i_{(\infty)} = i_p + i_n \rightarrow i_n = 0$ عند الاستقرار المستمر يعمل كسلك توصيل

$V_s = V_R + V_L \rightarrow V_L = iR + L\frac{di}{dt}$

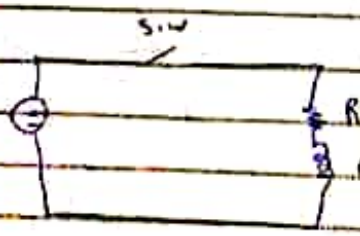
عند الاستقرار $i_p \rightarrow \frac{di}{dt} = 0$

$V_s = iR \rightarrow \boxed{\frac{V_s}{R} = i_p}$

$X_L = 2\pi f L \rightarrow f = 0 \rightarrow X_L = 0$

$\boxed{Z = R}$

2)



$i_{(\infty)} = i_p + i_n$

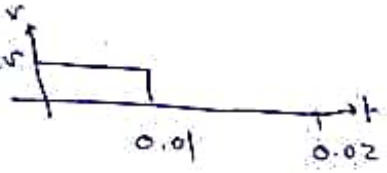
$\boxed{i_p = \frac{V_s}{R}}$

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Albalqa Applied University
Faculty of Engineering Technology

Student Name:
Student Number:

Q1/ A)



$$V_{avg} = \frac{1}{T}\int_0^{T_0} v_{in}\,dt$$

$$= \frac{1}{0.02}\int_0^{0.01} 2\,dt = \frac{2}{0.02}\left[t\right|_0^{0.01}$$

$$= \frac{2}{0.02}(0.01)$$

$$\boxed{V_{avg} = 1V}$$

1.5

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^{T_{on}} V^2\,dt}$$

$$= \sqrt{\frac{1}{0.02}\int_0^{0.01} (2)^2\,dt} \Rightarrow 2\sqrt{\frac{1}{0.02}(t|_0^{0.01}}$$

$$= 2\cdot\sqrt{\frac{0.01}{0.02}}$$

$$\boxed{V_{rms} = \frac{2}{\sqrt{2}} = \sqrt{2}\,V}$$

B)



$$V_{avg} = \frac{1}{T}\int_0^{t} v + \frac{1}{T}\int_{0.8}^{1} 1\,dt$$

$$V_{avg} = \frac{1}{1}\int_0^{0.5} 3\,dt + \frac{1}{1}\int_{0.8}^{1} 1\,dt$$

$$= 3(t|_0^{0.5}) + (t|_{0.8}^{1})$$

$$\boxed{V_{avg} = 1.5 + 0.2 = 1.7V}$$

1.5

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^{0.5} (3)^2\,dt + \frac{1}{T}\int_0^{0.8} (1)^2\,dt}$$

$$V_{rms} = \sqrt{9(t|_0^{0.5}) + (t|_0^{0.8}}$$

$$V_{rms} = \sqrt{4.5 + 0.8} = \boxed{\sqrt{5.3}\,V}$$

2

C)



$$V_{avg} = \frac{1}{0.3}\int_0^{0.2} 4\,dt + \frac{1}{0.3}\int_{0.2}^{0.3} (-1)\,dt$$

$$= \frac{10}{3}(t|_0^{0.2}) + \frac{10}{3}(-t|_{0.2}^{0.3}$$

$$= \frac{10}{3}\cdot 0.3 + \frac{10}{3}(-0.1)$$

$$= \frac{3}{3} - \frac{1}{3} = \frac{2}{3}V$$

$$\boxed{V_{avg} = \frac{7}{3}V}$$

V_{rms} for C

$$V_{rms} = \sqrt{\frac{1}{0.3}\int_0^{0.2} (4)^2\,dt + \frac{1}{0.3}\int_{0.2}^{0.3} (-1)^2\,dt}$$

$$= \sqrt{\frac{16}{0.3}(t|_0^{0.2}) + \frac{1}{0.3}(t|_{0.2}^{0.3}}$$

$$= \sqrt{\frac{16}{0.3}\cdot 0.2 + \frac{1}{0.3}\cdot 0.2}$$

$$= \sqrt{\frac{3.2}{3} + \frac{2}{3}}$$

$$\boxed{V_{rms} = \sqrt{\frac{34}{3}}\,V}$$

$$\boxed{V_{avg} = \frac{7}{3}V}$$

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2/ $R=20\,\Omega$ $V_{in}=400V$ $V_{ch}=2V$ $f=2KHz$ $d=0.4$

$$V_{avg}=\frac{1}{T}\int_0^{t_{on}}(V_{in}-V_{ch})\,dt$$

(constant)

$$=(400-2)\times(0.4)$$

$$V_{avg}=159.2V$$

$$T=\frac{1}{f}=0.5\times10^{-3}s$$

$$t_{on}=0.4\times0.5\times10^{-3}$$

$$t_{on}=0.2\,ms$$

$$V_{rms}=\sqrt{\frac{1}{T}\int_0^{t_{on}}(V_{in}-V_{ch})^2\,dt}$$

$$=\sqrt{\frac{1}{T}\int_0^{t_{on}}dt\times(V_{in}-V_{ch})^2}$$

$$=(400-2)\cdot\sqrt{\frac{1}{T}(t)_0^{t_{on}}}=\sqrt{d}$$

$$=398\times\sqrt{0.4}$$

$$V_{rms}=251.7V$$

$$R_1=\frac{R}{d}=\frac{20}{0.4}=50\,\Omega$$

$$\eta=\frac{P_{out}}{P_{in}}$$

$$\eta=\frac{3168}{3184}$$

$$=99.5\%$$

$$P_{out}=\frac{1}{T}\int_0^{t_{on}}\frac{V_o^2}{R}\,dt$$

$$=(V_{in}-V_{ch})^2\cdot\frac{d}{R}$$

$$=(398)^2\times\frac{0.4}{20}$$

$$=3168W$$

$$P_{in}=\frac{1}{T}\int_0^{t_{on}}V_{in}\cdot i\,dt \qquad i=\frac{V_{out}}{R}$$

$$=\frac{1}{T}\int_0^{t_{on}}V_{in}\frac{(V_{in}-V_{ch})}{R}\,dt$$

$$=\frac{400(400-2)}{20}\cdot\frac{1}{T}\int_0^{t_{on}}dt \Rightarrow d$$

$$=\frac{400\,(400-2)}{20}\times0.4$$

$$=3184W$$

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Faculty of Engineering Technology

Student Name:

Student Number:

First Exam, Second Semester: 2014/2015

Mechatronics Department

Course Title: Power Electronics	Date: 17/3/2015
Marks: 20	Time: 1 hour
Lecturers: Prof. Dr.Qazem M. Jabar	No. of pages: 4

Question 1: (2+2+1+1+1+1marks)

1-*Find the forced component of output instantaneous current with ac source , switch and RL load circuit after the switch is closed at t= 0*

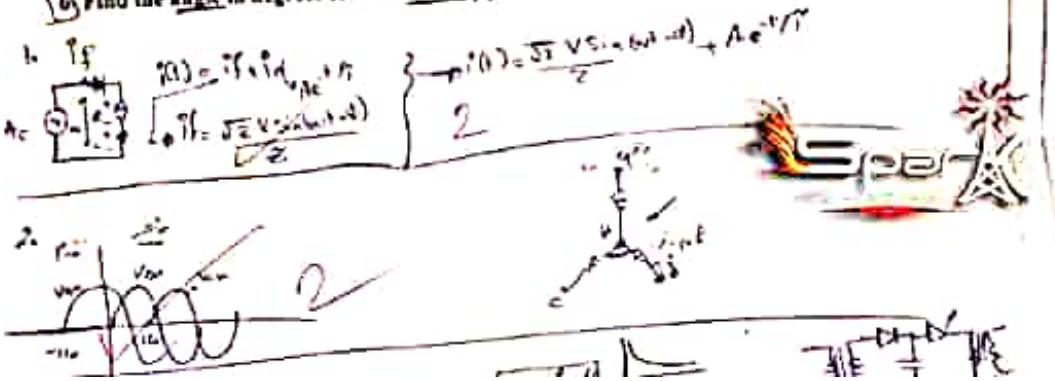
2-Draw the three phase waves of three phase source at the same curve with phase voltage vbn is reference voltage.

3-Write the relation ships of synchronizing firing signals (2signals) with syn phase voltage.

4-Find the forced component of output current of RC load circuit with switch and dc source after the switch is closed at t = 0.

5-count the three important parameters of power semi conductor switches.

6-Find the angle in degrees for time 0.008 [s].

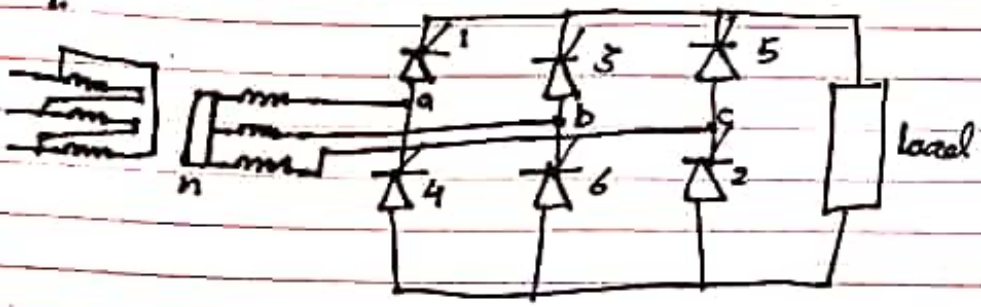


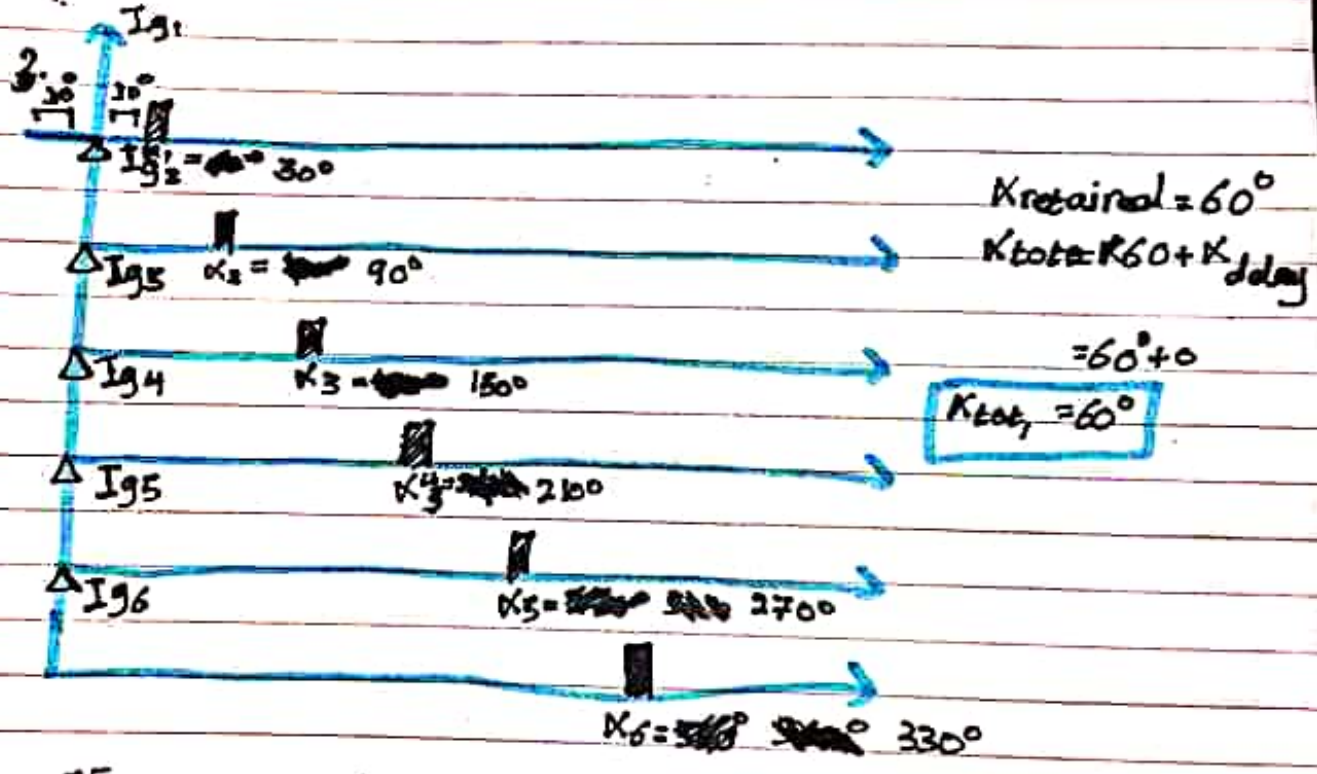
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#=Spark-Team

Question 1

1.





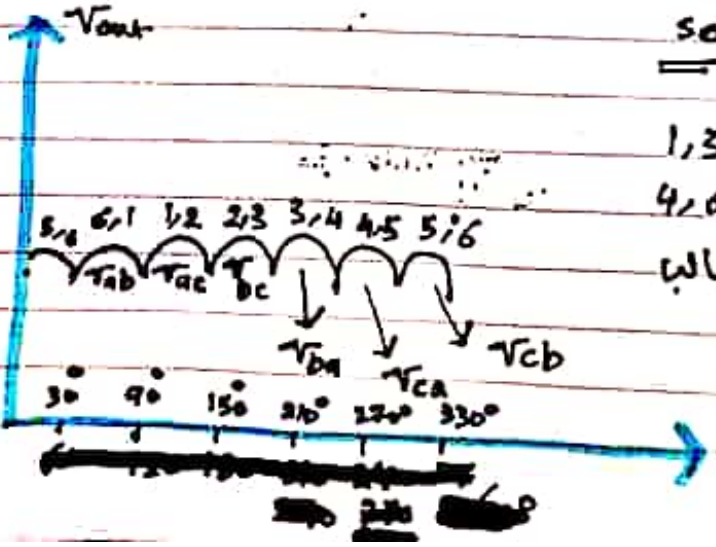
$K_{retained} = 60°$

$K_{total} = K_{60} + K_{delay}$

$= 60° + 0$

$K_{total} = 60°$

4.



← ارقام الثايرستورات حسب ال Sequence

← اسماء ال Voltages

1,3,5 الثايرستورات الموجبة

4,6,2 " السالبة

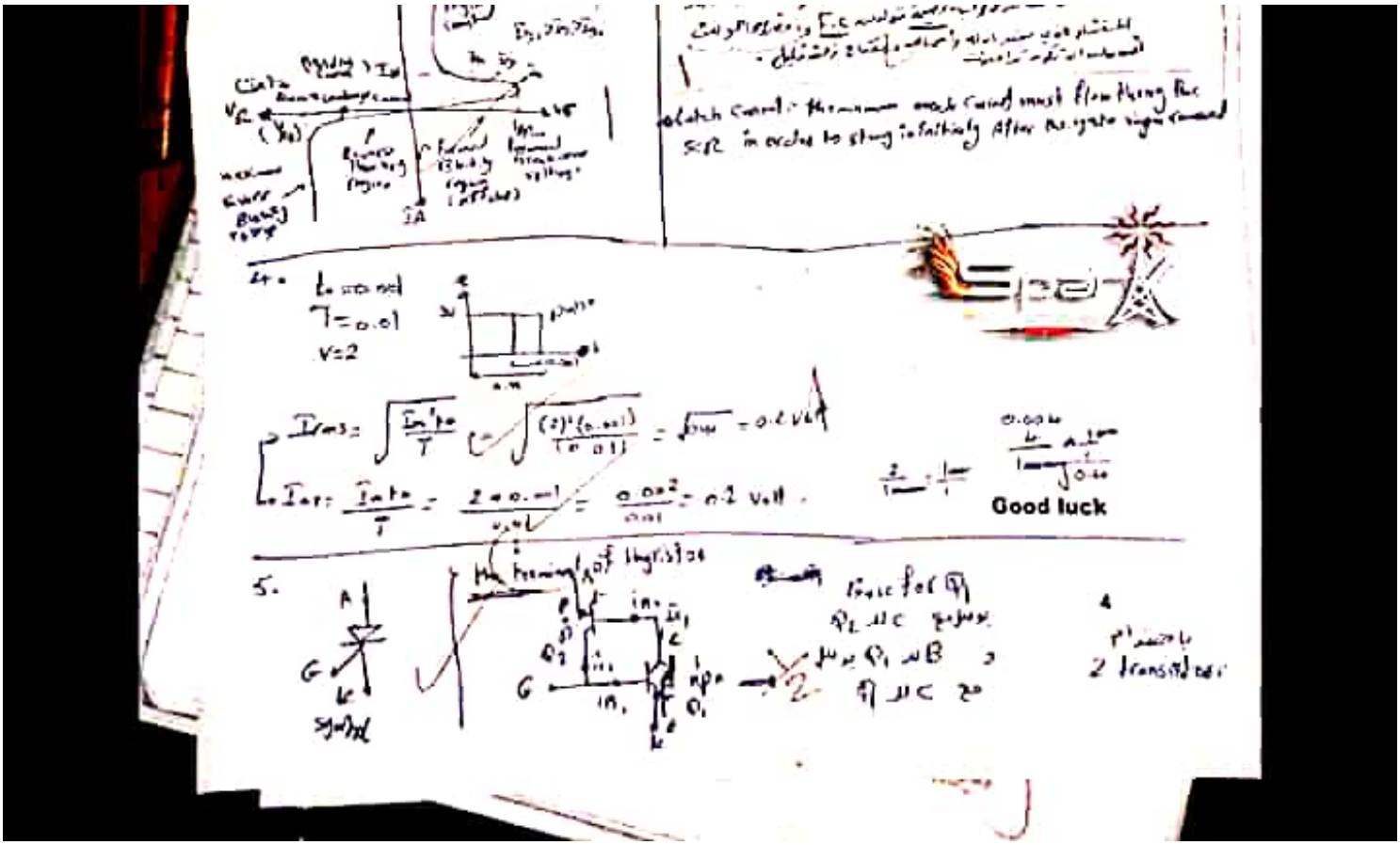
نبحط الموجب بالاول بعدين السالب

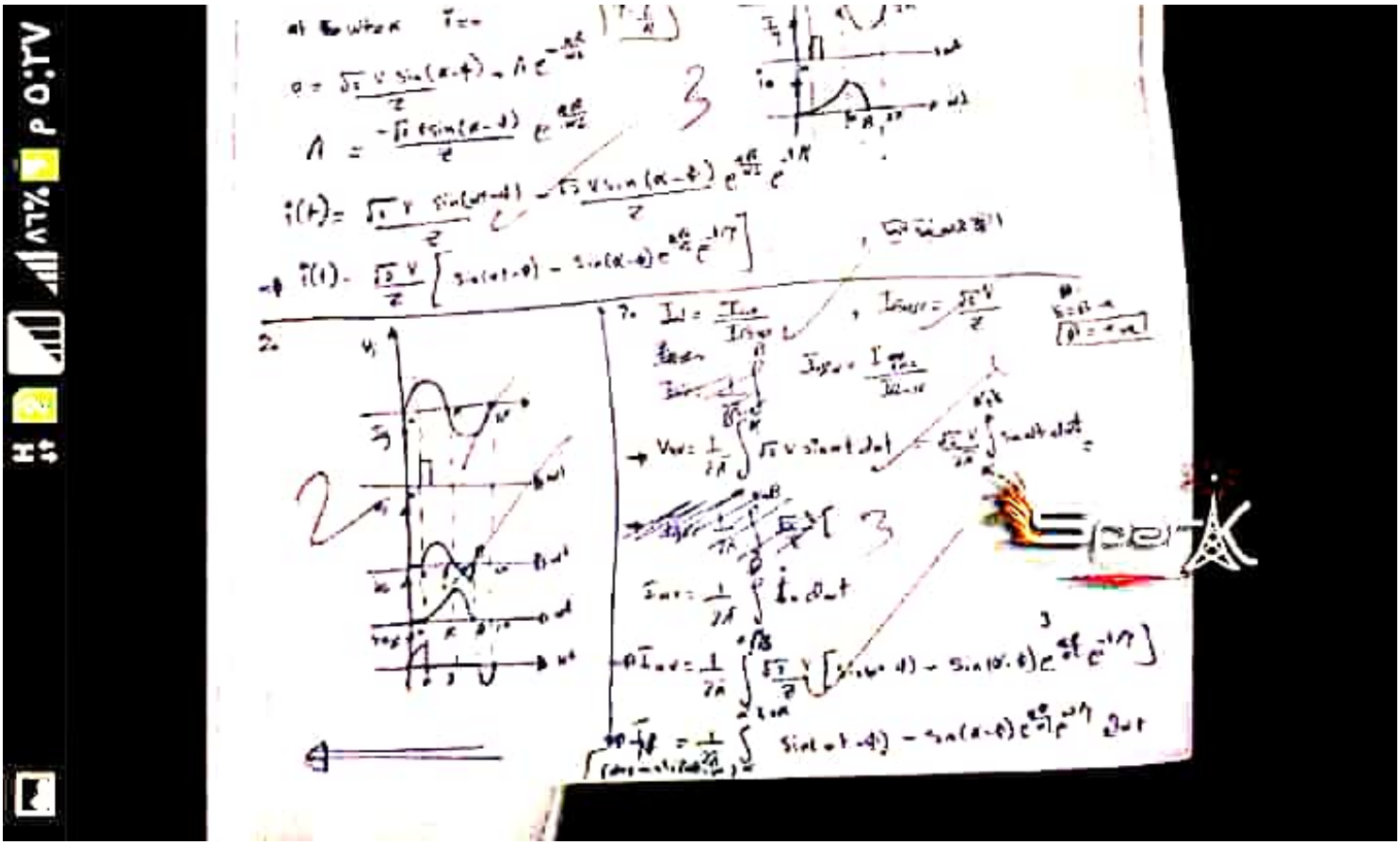
هنلاقي ⇦ 6/1

6/1 ← سالب هيكون b مع

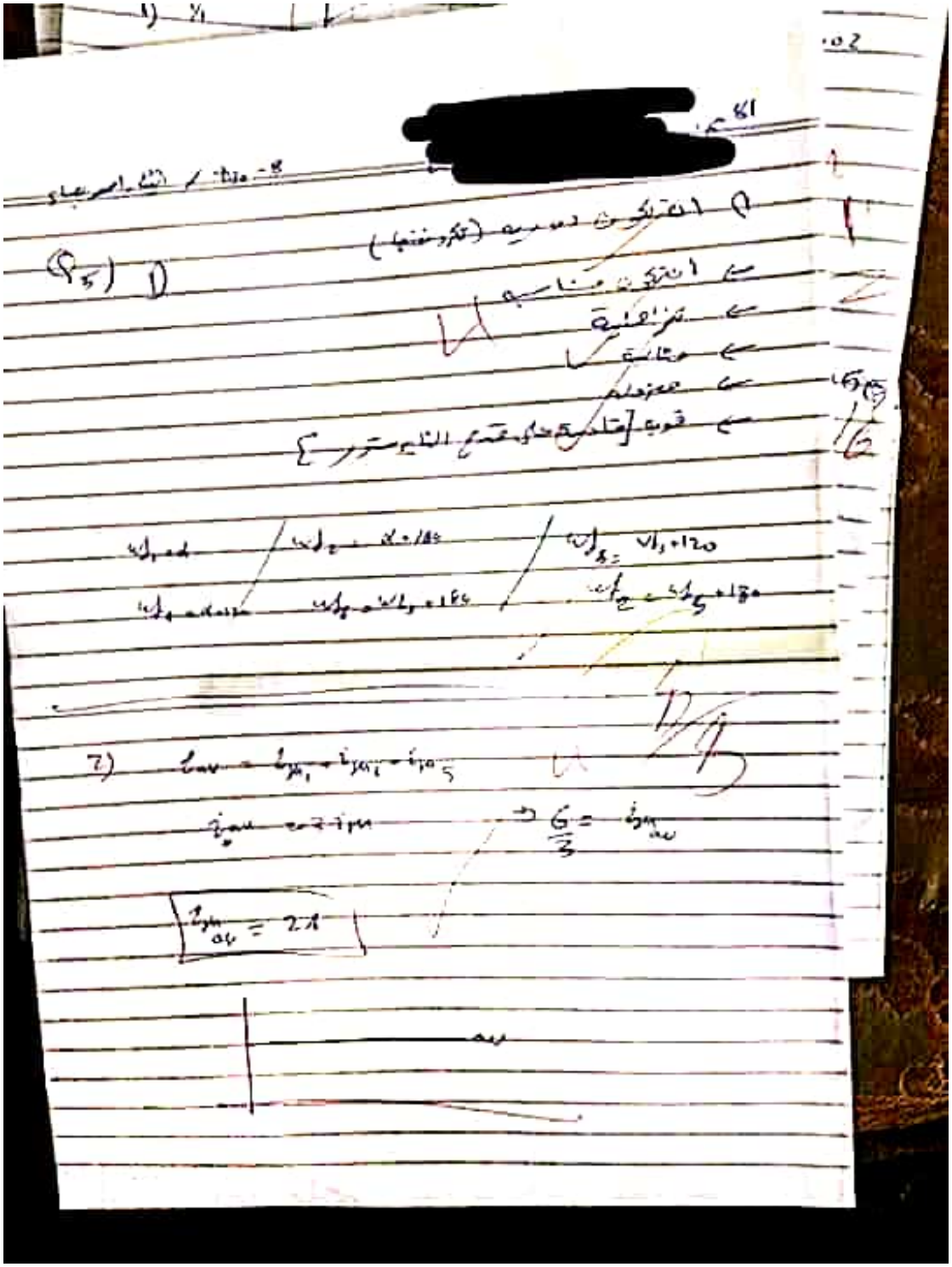
6/1 ← موجب هيكون a مع

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Latch current
Good luck
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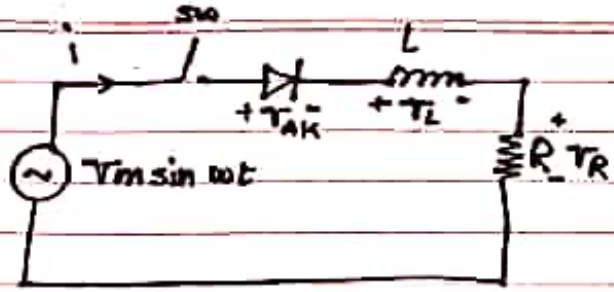


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Question



عند اللحظة X أو اللحظة B تكون قيمة التيار = صفر

عند بداية التشغيل $i=0$ ← عند B (نهاية التشغيل) $i=0$



at $t=X \rightarrow$ sw is on

$$-V_m \sin \omega t + V_L + V_R = 0$$

$$V_m \sin \omega t = V_L + V_R$$

$$V_m \sin \omega t = L\frac{di}{dt} + iR$$

if $\Rightarrow$ when $\frac{di}{dt} = 0$

$$i_F = \frac{V_m \sin(\omega t - \phi)}{R}$$

source is off

$$i_N \Rightarrow 0 = L\frac{di}{dt} + iR$$

$$i_N = A e^{-t\frac{R}{L}}$$

$$i_{tot} = \frac{V_m \sin(\omega t - \phi)}{R} + A e^{-t\frac{R}{L}}$$

A $\Rightarrow$ from initial condition $\Rightarrow$ sw $t=0 \Rightarrow i=0$

$$0 = \frac{V_m \sin(-\phi)}{R} + A$$

$$A = \frac{V_m \sin\phi}{R}$$

$$i_{tot} = \frac{V_m \sin(\omega t - \phi)}{R} + \frac{V_m}{R} \sin\phi \; e^{-t\frac{R}{L}}$$

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40

[redacted]

السبت - 1/ربيع / 9:30 - ص

Q:1

1) Y1

$f_{(t)} = t+2 \quad 0 < t < 0,02$

2/2

$0 < t \leq 0,3(s) \rightarrow f(t) = \text{const}$

$0,35 < t < 0,4(s) \rightarrow f(t) = \text{const}$

30

0,015

$f(t) = \cos(t) \quad 0 < t < 0,015$

(5/6)

3)

$I_{av} = \frac{I_m t_0}{T} = \frac{10 \times 0,1}{0,4} = 2,5\ A$

$I_{rms} = \sqrt{\frac{I_m^2 t_0}{T}} = \frac{(10)^2 \times 0,1}{T} = 5\ A$

3)

$V_{av} = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_{(t)}\, dt$

$V_{rms} = \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\beta} V_{(t)}^2\, dt}$

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مهم جداً

=> Draw V_{L-L} voltages consider that V_{bn} is reFFerence Voltage.

=> reFFerence يبساطة معنى كلمة
يعني انه الـ (phase shiFt) Zero
=> فبتعبير V_{bn} هي المرجعية
$V_{bn} = V_m \sin(\omega t - 0)$
طبعاً احنا بنحسبش +sequence
يعني
a → b → c
بما انه b هي المرجعية
b → c → a

phase Voltages:
$V_{pn} = V_m \sin \omega t$
$V_{cn} = V_m \sin(\omega t - 120)$
$V_{an} = V_m \sin(\omega t + 120)$ $-240°$

لحساب الـ V_{L-L}
$V_{ab} = V_{an} - V_{bn}$
$= V_m \sin(\omega t + 120) - V_m \sin(\omega t - 0)$
$= V_m \angle 120° - V_m \angle 0°$
$= V_m (\angle 120° - \angle 0°)$
$= V_{bn} \angle 120° - \angle 0°) \Rightarrow$ هنا بعوض بدل V_m بـ V_{bn} (هي نفسها V_{bn}) على الطريقة الحسابية
$= V_{bn} (\sqrt{3} \angle +150°)$

$\boxed{V_{ab} = \sqrt{3} V_{bn} (\sin(\omega t + 150°))}$ ← 150° لقدمت بمقدار

$\boxed{V_{bc} = \sqrt{3} V_{bn} \sin(\omega t + 30°)}$ ← V_{bc} 30° لقدمت بمقدار

$\boxed{V_{ca} = \sqrt{3} V_{bn} \sin(\omega t - 90)}$ ← -90° تأخرت بمقدار

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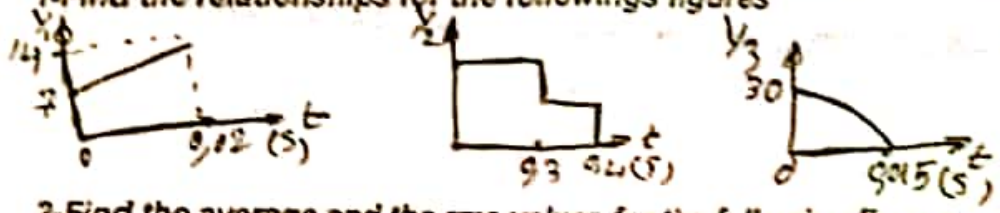
Faculty of Engineering Technology Student Number:

Mid Exam, First Semester: 2017/2018
Mechatronics Department

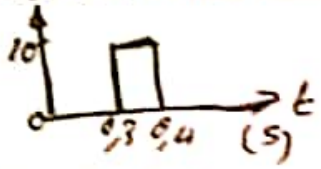
Course Title: Power Electronics	Date: 19/11/2017
Marks: 40	Time: 1:25 hour
Lecturers: Prof. Dr.Qazem M. Jaber	No. of pages: 1

Question 1: *(6marks)*

1-Find the relationships for the followings figures



2-Find the average and the rms values for the following figures



3-Write the expressions for the rms and the average values of the instantaneous output voltage with one period.

Question2: *(4marks)*

1-Find the current in the closed circuit with dc source and RL load circuit
2-Find the current in the circuit with switch ,dc source, and RL load circuit afte the switch is closed.

Question3: *(12marks)*

1-Draw the power circuit of three phase half wave controlled rectifier with transformer, R load circuit and $\alpha = 0^{\circ}$.2-Draw the all curves versus time that explain the work of the power circuit in item 1.3-Draw the anode cathode voltage of thyristor that is connected with bn phase in the circuit is shown in item 1

Question4: *(16marks)*

1-Derive the out put instantaneous current of the circuit contains ac source, thyristor with $\alpha = 60^{\circ}$ and RL load circuit
2-Draw the practical circuit contains relay and the singe phase half controlled rectifier and explain the work with curves
3- For the circuit in the item 1with220rms phase voltage and R=XL =5Ω calculate: Iav , Irms ,normalized values, β and power factor

Question5: *(8marks)*

1-Count the technical specification of the firing circuit for three phase full wave controlled rectifier.
2-If the average output current is 6 amps find the thyristor current and explain it.

Good lu

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③ $\eta = \frac{P_{dc}}{P_{ac}}$

$= \frac{V_{av} I_{av}}{V_{rms} I_{rms}}$

$= \frac{I_{av}^2 R}{I_{rms}^2 R}$

$= \sqrt{I}$

$\eta = \left(\frac{I_{av\,load}}{I_{rms\,load}}\right)^2$

تقويض مباشر

تقويض جانبي

④ $pIV = \text{sen}\theta . |V_m|$

$= \sqrt{2} * 110\ V$

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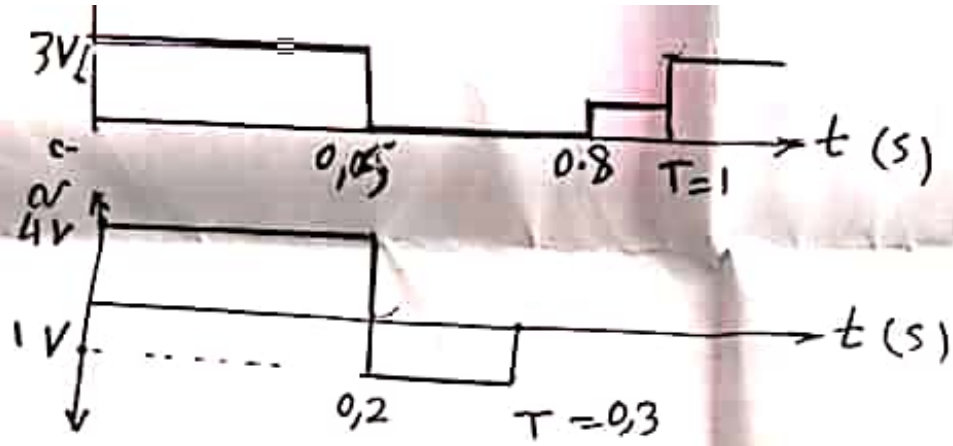


Fig.1

2-A step –dawn dc chopper has a resistive load of R = 20Ω and input voltage of V = 400 V. When chopper is ON, its voltage drop is2 V and the chopping frequency is 2 kHz. If the duty cycle is 40%, determine

1 *–Average output voltage*
2 *–RMS value of output voltage*
3 *–Effective input resistance of chopper*
4 *–Chopper efficiency.*

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111. The electrical breakdown strength of insulating materials depends on

(A) value of applied voltage

(B) impurities in dielectric material

(C) pressure, temperature and humidity

(D) all of the above

112. Which of the following gas has been used as insulating medium in electrical appliances ?

(A) Nitrogen

(B) Carbon dioxide

(C) Sulphur hexafluoride

(D) Freon

113. Electro mechanical breakdown of solid insulating materials occurs due to

(A) magnetic force

(B) vibrations

(C) mechanical stresses produced by the electrical field

(D) electrical stresses produced by the voltage fluctuations.

114. Surge voltages originate in power systems due to

(A) lightning

(B) switching operations

(C) faults

(D) any of the above

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Albalqa Applied University
Faculty of Engineering Technology

Student Name:
Student Number:

Mid Exam, Summer Semester: 2017/2018
Mechatronics Department

Course Title: Power Electronics	Date: 19/07/2018
Marks:40	Time: 1.15 hour
Lecturers: Prof. Dr.Qazem M. Jaber	No. of pages: 1

Question 1: *(12marks)*

1-Count the technical specifications of the dc and the ac sources
2-Count he difference between the real and the not real sources
3-Replace the thyristor with α =30□ by diode.

Question2: *(10marks)*

1-Find the current in the circuit contains dc source, rl load circuit and switch, where the switch has been closed
2-Find the current in the circuit with dc source and the rc load circuit with switch, after the switch is closed at t = 0.

Question3: *(10marks)*

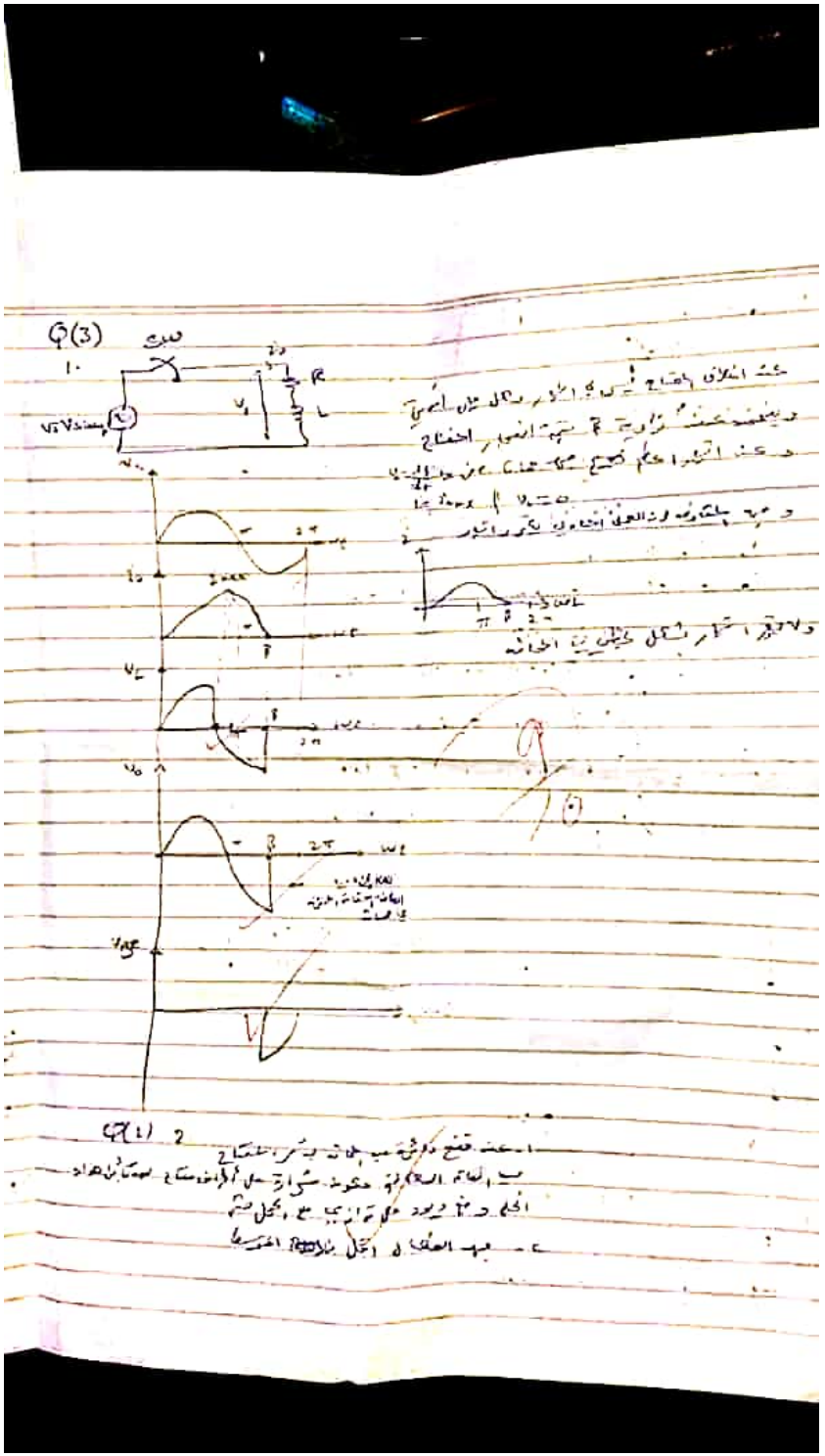
1-Explain the quality work of the circuit contains ac source, switch and the rl load circuit after the switch is closed at t = 0
2-Count the disadvantage the circuit in item 1 and explain how to eliminate it

Question4: *(8marks)*

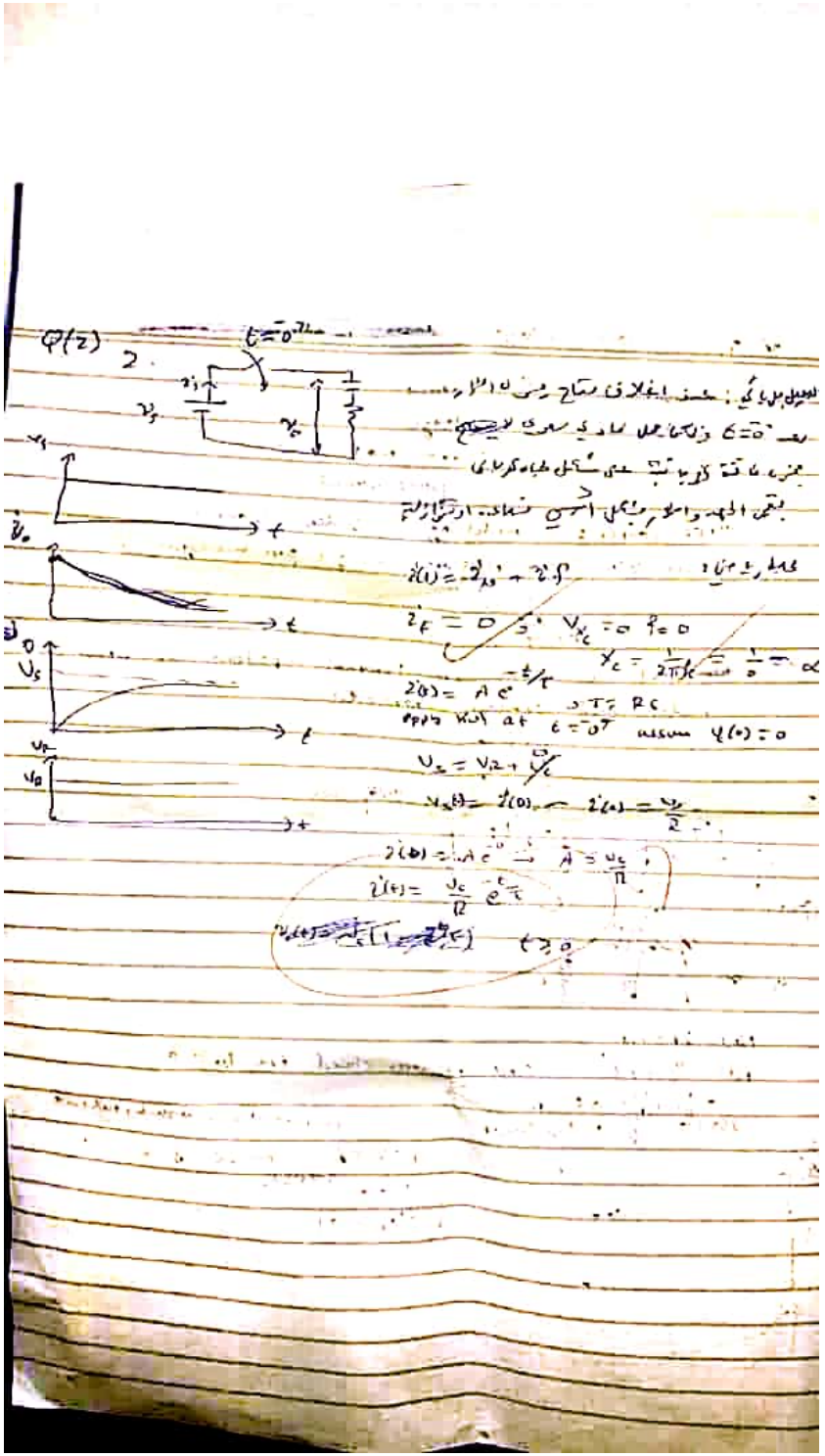
1-Find the output power of the r = 4Ω and E =50V are connected with switch, diode and 220 rms value source after the switch is closed at t = 0.
2-Draw the instantaneous output voltage and the instantaneous output current versus time.

Good luck

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